

Tropical college of medicine



Obstetrics and gynecology Nursing II for 3rd year Bsc nursing student

Dr Abreham Tizazu(MD)
2015 EC
Debremarkos,Ethiopia

Premature rupture of membranes

- Membrane rupture that occurs spontaneously before the onset of labor
 - ⇒ Term PROM: Rupture of membranes after 37 weeks
 - ⇒ Preterm PROM: Before 37 weeks
- Prolonged PROM: Longer 12 hours
- Abc started after 8 hours if no labor started

Risk factors

- Mechanical factors: Multifetal gestation, polyhydramnios, pulmonary diseases, cervical conization/ LEEP/ cerclage.
- Urogenital infections: UTI, cervicitis, GBS, bacterial vaginosis.
- Previous history of preterm PROM, preterm labor.
- 2nd and 3rd trimester bleeding (e.g. abruption placenta).
- Other risk factors: Low socioeconomic status, nutritional deficiencies, low BMI, smoking and connective tissue disorders

PROM diagnosis

- History

- ⇒ Classic presentation is a sudden "gush" of clear or pale yellow vagina fluid.

- ⇒ Complications of PROM infection, preterm labor, cord prolapse

PROM diagnosis...

- Physical findings
 - ⇒ Negative discrepancy
 - ⇒ If complicated, uterine contraction, tenderness
 - ⇒ Speculum exam with or without Valsalva maneuver
 - ⇒ Best confirmatory methods direct observation of amniotic fluid

PROM diagnosis...

- Ferning test /arborized crystals
- Pad test
- Nitrazine paper test
- Dye amnioinfusion - golden
- Ultrasound: support diagnosis & fetal wellbeing.
- CBC
- U/A, culture & sensitivity

Complications

Maternal risks	• Infection of the amniotic cavity • Placental abruption • Higher risk of cesarean delivery
Fetal	• Infection • Fetal distress • Umbilical cord compression • Umbilical cord prolapse • Fetal death
Neonatal infection	• Acute congenital pneumonia • Sepsis • Meningitis • Late onset bacterial or fungal infections

Treatment

- Indications for expedite delivery
 - ⇒ Onset of labor
 - ⇒ Gestation age ≥ 37 wks
 - ⇒ Evidence for nonreassuring fetal status
 - ⇒ Evidence for chorioamnionitis
 - ⇒ Lethal congenital anomalies
 - ⇒ Intrauterine fetal death
 - ⇒ High risk of cord prolapse and abruption placenta
- Expectant management for gestational is below 34 weeks and both the fetal and maternal conditions are stable.

Treatment

- Expectant management

- ⇒ Admit to the ward

- ⇒ Avoid digital cervical (pelvic) examination.

- ⇒ Advise bed-rest

- ⇒ Dexamethasone 6 mg IM BID for 48hrs for lung maturity

- ⇒ Antibiotics

1. Ampicillin 2gm IV QID and Erythromycin 250 mg P.O QID for 48 hours then
2. Amoxicillin 500 mg P.O TID & Erythromycin 250 mg. P.O QID for 5 days

Unit-5:Operative delivery

1

- Forceps delivery

2

- Vacuum delivery

3

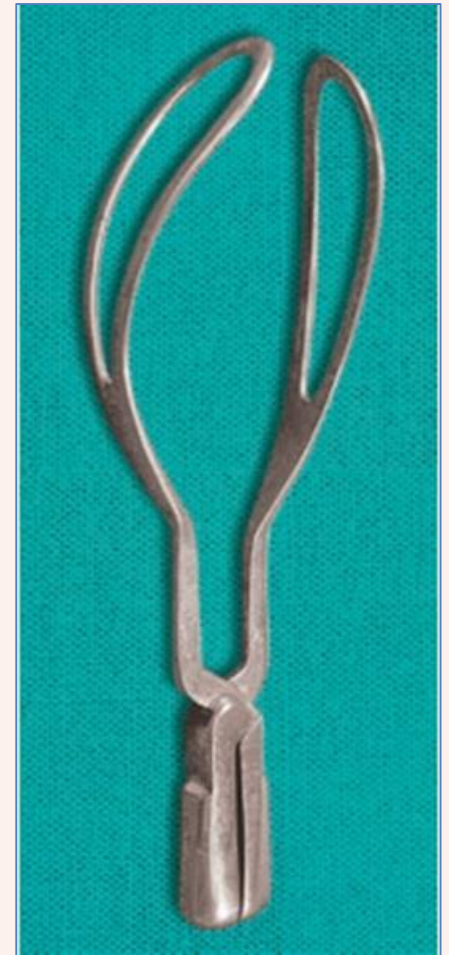
- Cesarean section

Objective

- Define each operative delivery
- Describe types of operative deliveries.
- Identify indications & contraindication for operative deliveries.
- Explain prerequisite condition operative deliveries.
- Describe complication of operative deliveries.
- Describe definition, indication method and complication of induction

Obstetric forceps

- Is a pair of instruments specially designed to assist extraction of fetal head and thereby accomplishing delivery of the fetus.
- Forceps functions
 - ⇒ Traction
 - ⇒ Rotation.



Anatomy of obstetric forceps

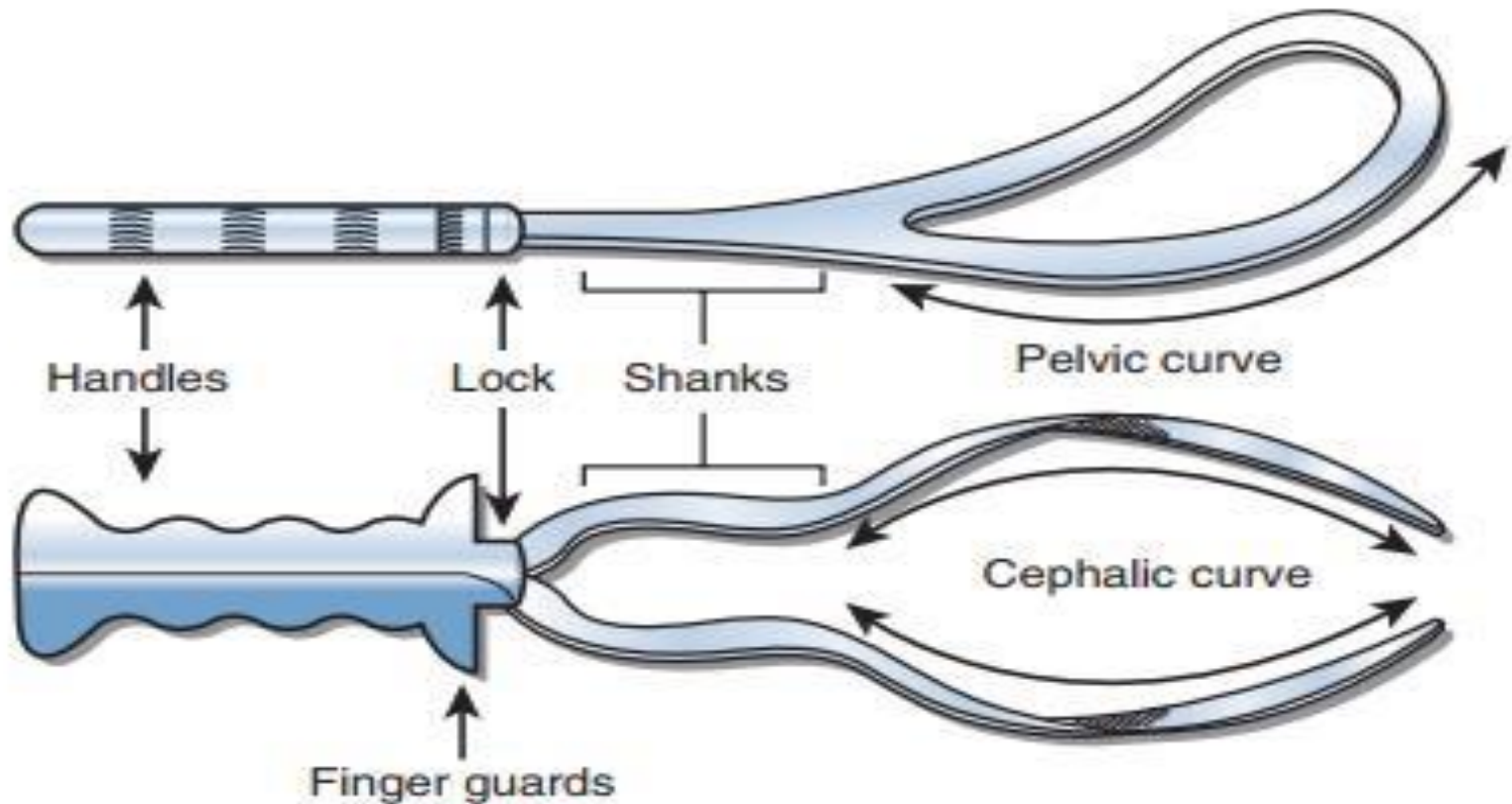


FIG 14-1 Anatomy of the forceps.

Type of forceps

Based on function

Classic
forceps

Rotational
forceps

Specialized
forceps

Based on application level

Outlet
forceps

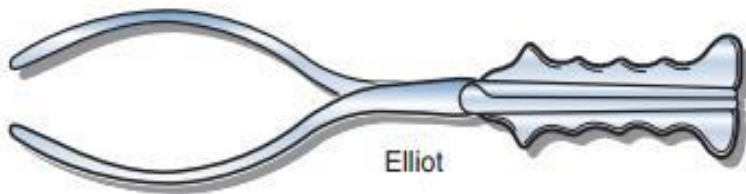
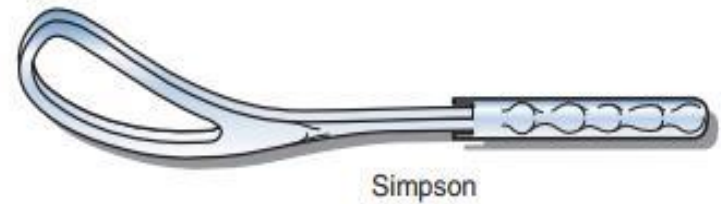
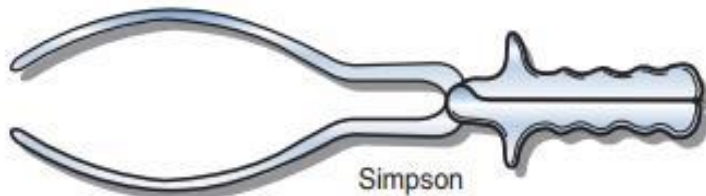
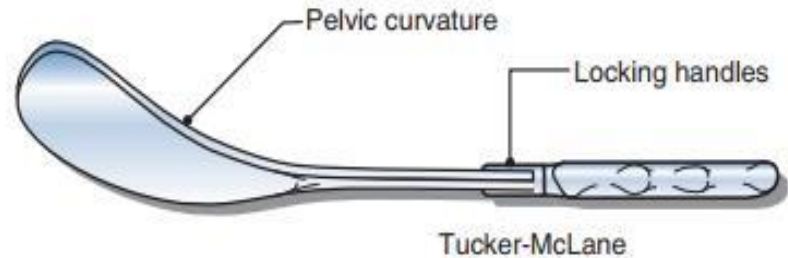
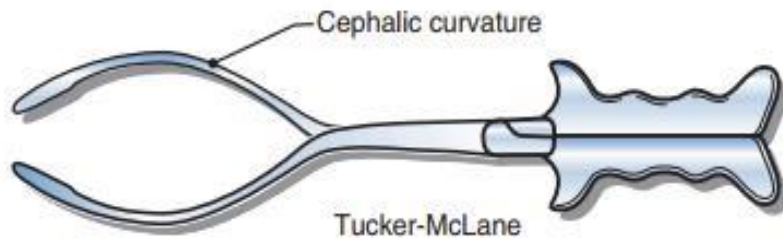
Low
forceps

Mid
forceps

Classic forceps

- Is conventional traction forceps
- Are typically used when rotation of the vertex is not required.
 - ⇒ Simpson forceps (has parallel shanks)
 - ⇒ Elliot forceps (has overlapping shanks)
 - ⇒ Tucker-McLane forceps (has solid blades)

Classic forceps....



Rotational forceps

- Used for rotation
- Absence of a pelvic curve facilitates rotation of the vertex.
- Cannot be depressed sufficiently low against the perineum.
 - ⇒ Tucker-McLane
 - ⇒ Kielland

Rotational forceps.....

② Rotational forceps



Fig. 36.15 : Kielland's Forceps

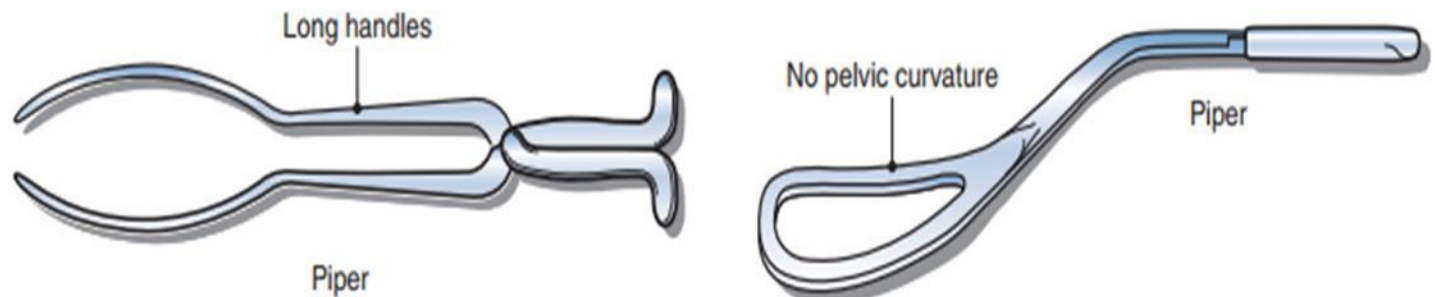
Piper forceps is a specialized forceps, used to assist the delivery of the after-coming head of breech. It has a cephalic curve, reverse pelvic curve, long parallel shanks that permit the baby's body to rest against it during head delivery.

Specialized forceps

- Forceps to assist with delivery of the after coming head during vaginal breech delivery.

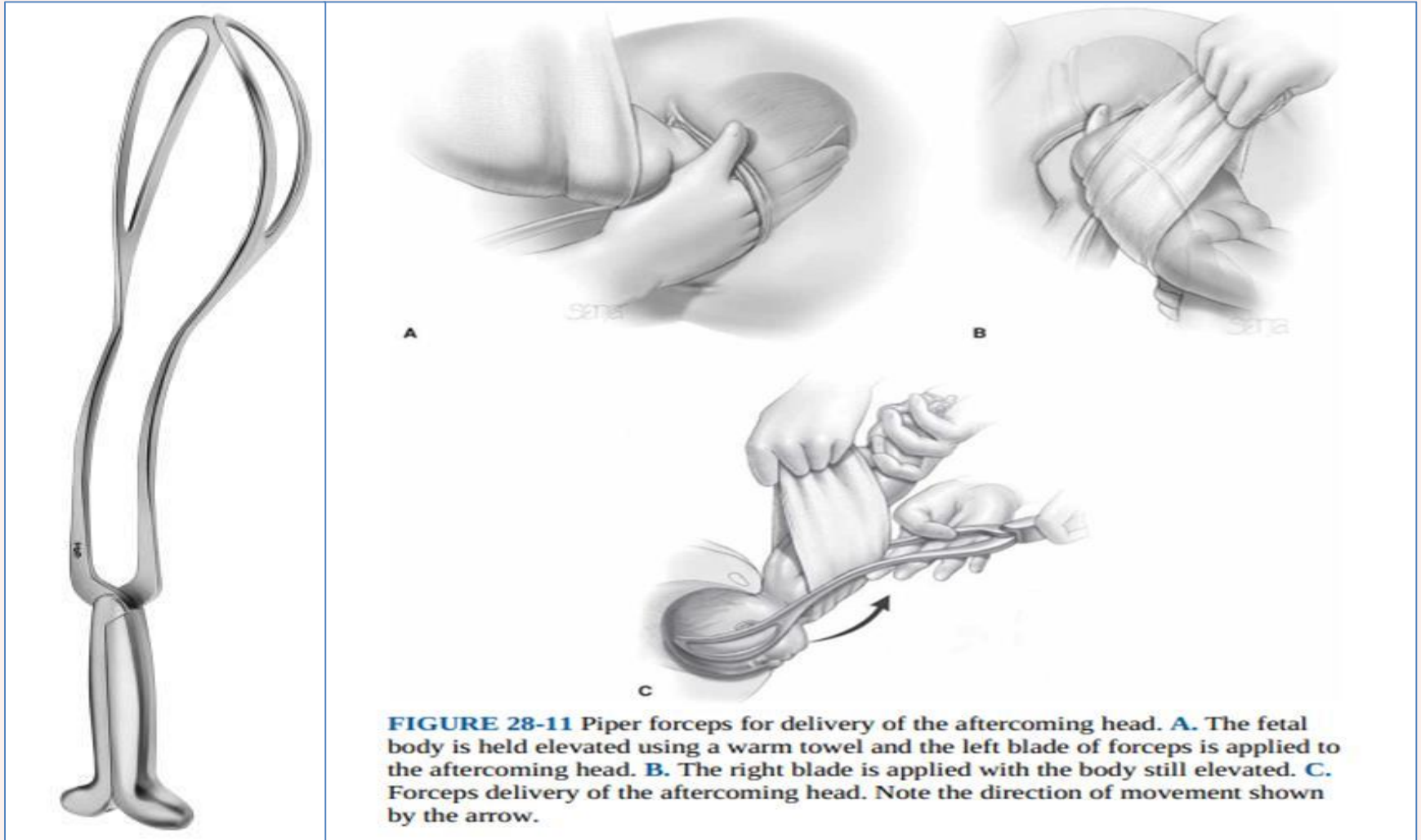
⇒ Piper forceps

③ Forceps for delivery of aftercoming head of the breech



JWKOI *M. Coley*

Specialized forceps...



Forceps classification

1	Outlet	• Applied when the fetal head is at station +3 (at the pelvic floor). • Scalp is visible at the introitus without separating the labia. • Sagittal suture is in AP diameter or right or left OA or OP. • Fetal head is at or on perineum. • Rotation does not exceed 45 degrees.
2	Low	• Applied when the station is +2 or below not on pelvic floor • Rotation ≤ 45 degrees to LOA/ROA to OA or. • Rotation ≥ 45 degrees to LOP/ ROP to OP.
3	Mid forceps	• Station is above +2 cm but head is engaged. • Station is between 0 and till 2cm. • Done $\Rightarrow$ Only with skilled operator. $\Rightarrow$ When assessment suggests a high chance of success.

Indication and contraindications of forceps

Indication	Contraindications
• Prolonged second stage • Fetal compromise in 2nd stage of labor ⇒ Non reassuring fetal HR ⇒ Cord prolapse ⇒ Abruptio placenta • After coming head of breech • Shortening of the 2nd stage of labor ⇒ Maternal exhaustion ⇒ Maternal cardiopulmonary ⇒ Cerebrovascular disease ⇒ Proliferative retinopathy ⇒ Severe anemia ⇒ Pregnancy induced hypertension ⇒ Eclampsia	• Absence of full dilatation of cervix. • Cephalopelvic disproportion. • High station of fetal head • If uterine contraction cease • Lack of experience of operator. • Mentum posterior face presentation. • Hydrocephalic infant • Brow presentation.

Prerequisites for forceps application

- Presentation & position(Vertex⇒OA,Face⇒mento-anterior (MA)).
- Station of +2 or below
- Fully dilated cervix
- Ruptured membranes
- Adequate pelvis
- Provider should be skilled in performing forceps delivery
- No contraindication to vaginal delivery

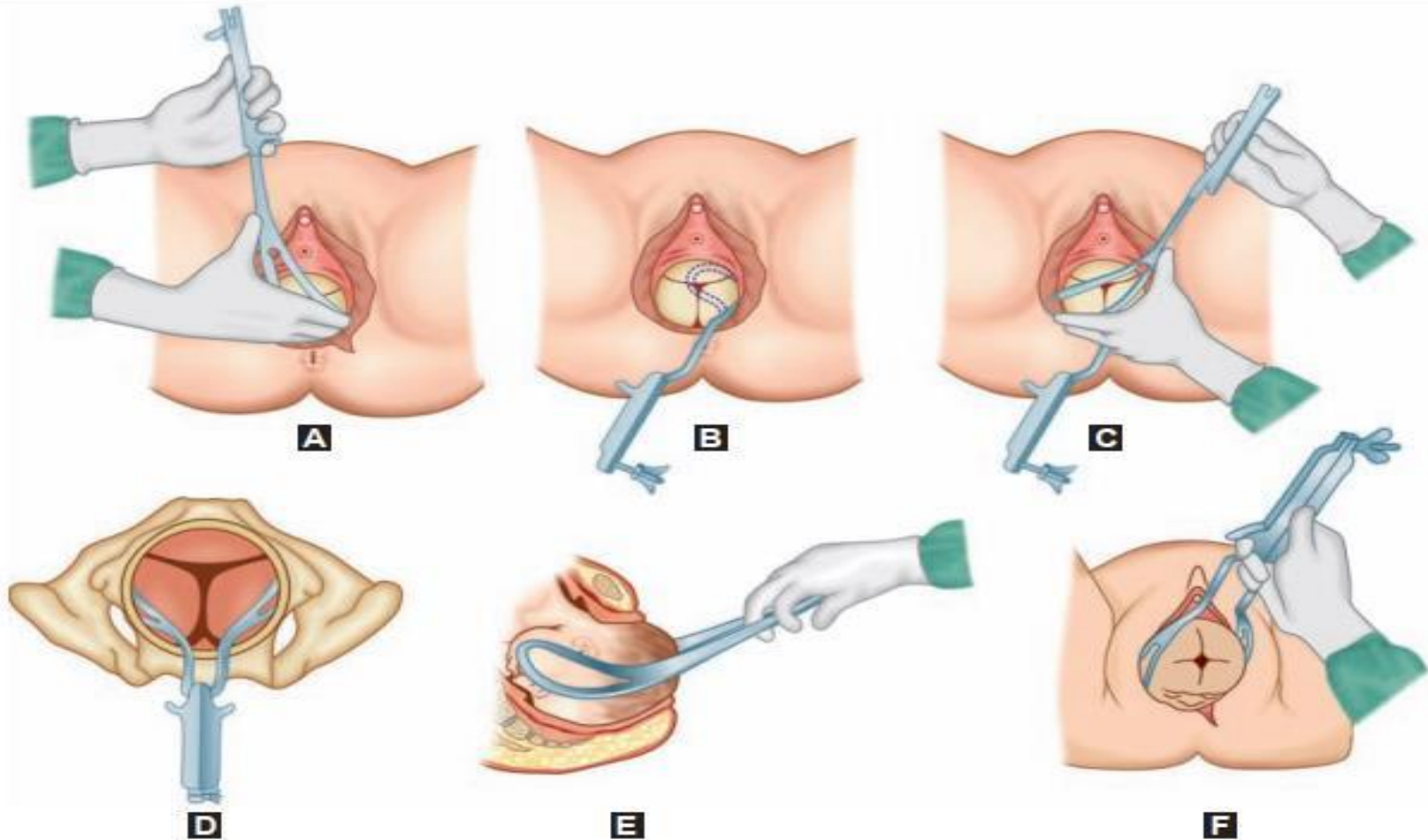
Preparation

- Counsel and get written consent.
- Empty bladder.
- Local anesthesia infiltration for episiotomy (if required).
- Prepare all necessary equipment and asses functionality

Procedure

1. Completely locked forceps in front of the perineum to orient and identify the right and left blades.
2. Insert two fingers of the right hand into the vagina
3. Slide the left blade gently between the head and fingers to rest on the side of the head(repeat the same maneuver on right side)
4. Check that application and no maternal tissue is entrapped.
5. After locking, apply steady traction

Procedure



Figs 36.12A to F: Steps of low forceps operation — (A) Introduction of the left blade; (B) The handle lying flat on the perineum after introduction; (C) Introduction of the right blade; (D) Showing perfect apposition and locking of the blades; (E) Bimalar, biparietal placement of blades and position of the fingers during traction; (F) Change in the grip in the final stage of delivery

Procedure...

6. After locking, apply steady traction
 - Between contractions check FHB and application of forceps.
 - When the head crowns, make an episiotomy if necessary.



Fig. 36.13: The blades are equidistant from the lambdoid suture indicating correct application

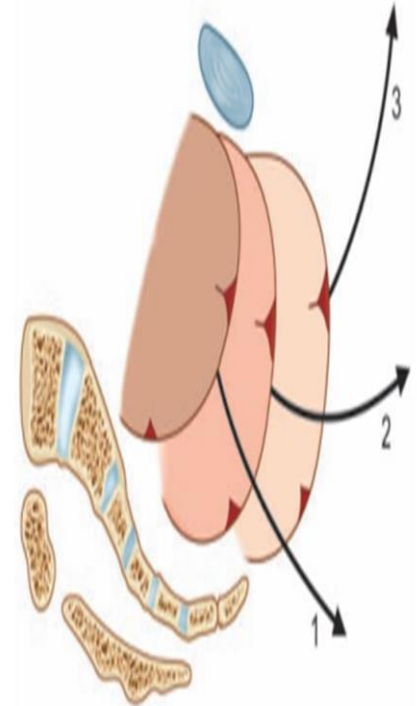


Fig. 36.14: Showing direction of traction along the axis of the birth canal — (1) Downwards and backwards; (2) Straight horizontal pull (3) Upwards and forwards

Procedure...

7. Remove first the right forceps followed by the left.

- Further care

⇒ After delivery inspect the vagina and cervix; and repair any tear or episiotomy.

⇒ Examine the newborn .

⇒

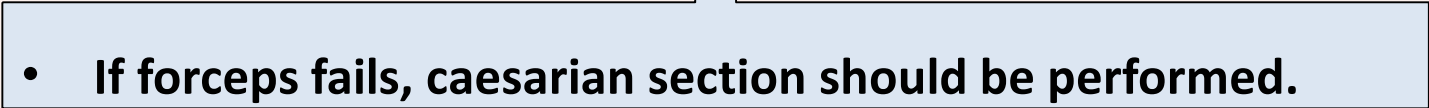
Forceps



FIGURE 29-11 Branches are removed in the opposite order from that in which they were originally placed. The fingers of the right hand, covered by a sterile towel, bolster the perineum. The thumb is placed directly on the head to prevent sudden egress. (Reproduced with permission from Yeomans ER: Operative vaginal delivery. In Yeomans ER, Hoffman BL, Gilstrap LC III, et al (eds): Cunningham and Gilstrap's Operative Obstetrics, 3rd ed. New York, McGraw-Hill Education, 2017.)

Failed forceps

- When a deliberate attempt in vaginal delivery with forceps has failed to expedite the process, it is called failed forceps
- Forceps Failed If
 1. Locking of forceps is difficult
 2. Fetal head does not advance with each pull.
 3. Fetus is undelivered after three pulls with no descent
 4. Fetus is undelivered after 30minute

- 
- **If forceps fails, caesarian section should be performed.**

Advantages over vacuum

- Fast
- Active rotation possible
- Traction independent of contraction
- Also used in face (MA) & after coming head of breech
- Can be used for preterm(≤ 34 weeks)
- Less failure to effect delivery
- Less fetal trauma

Complication of forceps delivery

Fetal complications	Maternal complications:
• Injury to facial nerves • Lacerations of the face and scalp • Cephalhematoma • Intracranial hemorrhage. • Fracture of the facial bones or skull	• Tear of cervix, vagina, or vulva. • Uterine rupture • Postpartum hemorrhage (traumatic) • Shock • Infection • Delayed or long-term sequel ⇒ Chronic low backache ⇒ Genital prolapsed ⇒ Incontinence.

Complication of forceps delivery...



FIGURE 33-4 Left facial nerve injury. This was almost completely resolved two days after delivery.

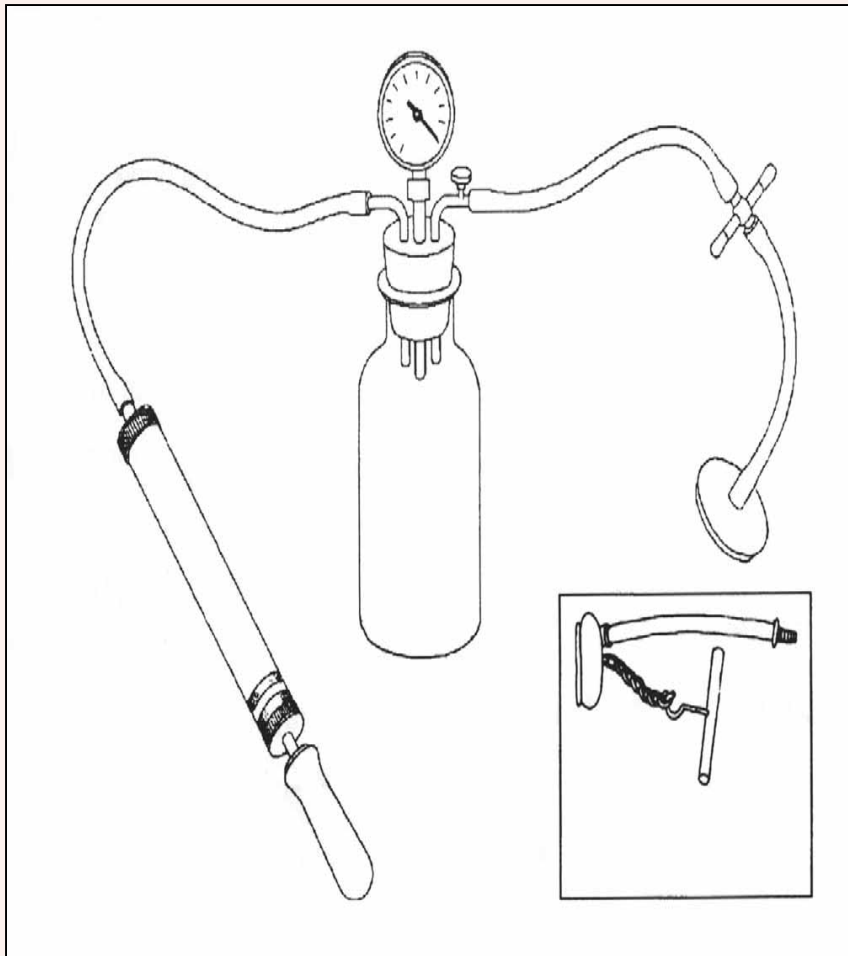


Vacuum extraction (ventouse)

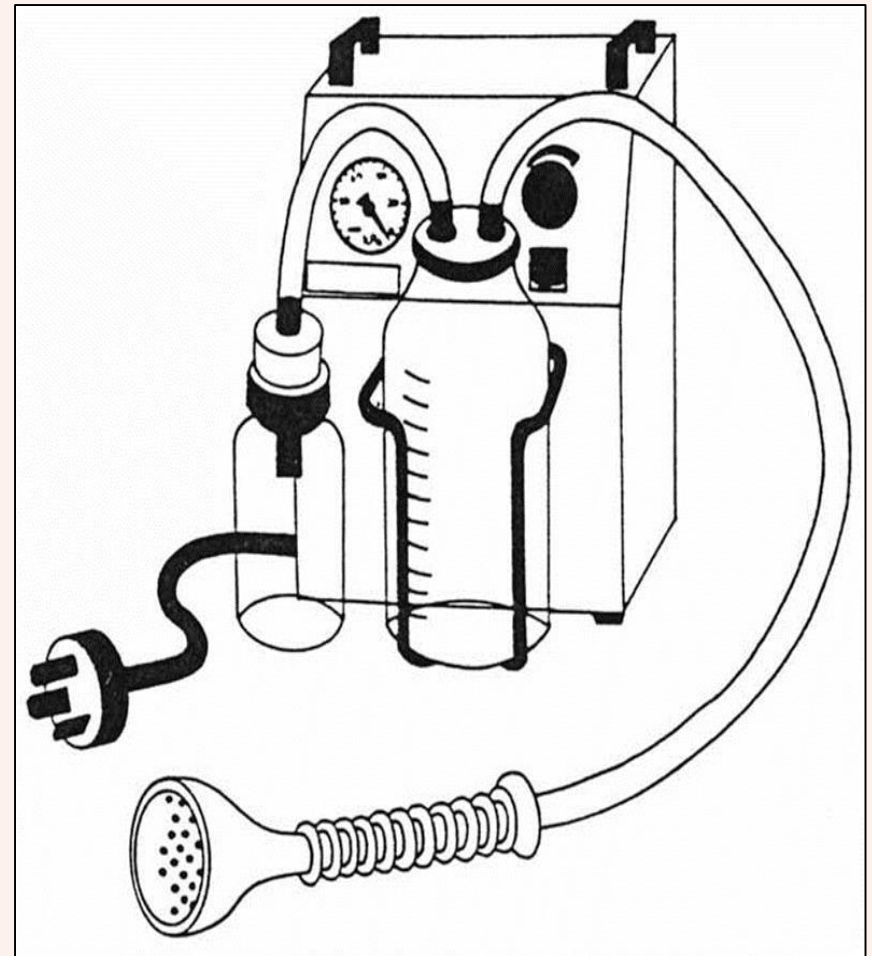
- It is an instrumental device designed to assist delivery by creating a vacuum between it and the fetal scalp.
- Function of the cup
 - ⇒ Flexion
 - ⇒ Descent
 - ⇒ Autorotation



Types of vacuum extractors



Vacuum hand pump



Electrical vacuum pump with soft cup

Types of vacuum extractors...

Cup type

Metal cup/M-style

- Rigid
- Has more traction force
- Size: 40, 50, & 60 mm
- Can be autoclaved
- Is a mushroom-shaped
- Have a higher success rate
- Easier cup placement in OPP
- ↑ risk of fetal scalp injuries.
- More difficult to apply

Soft cup/Pliable

- Vacuum attained more quickly
- Are bell, mushroom or funnel shaped.
- Fewer neonatal scalp injuries.
- Disposable and reusable
- Faster and easy application
- Less pronounced chignon
- Fewer superficial scalp injuries
- Less intracranial hemorrhage
- Autorotation
- Has higher failure rate
- Easy to apply.

Types of vacuum extractors...

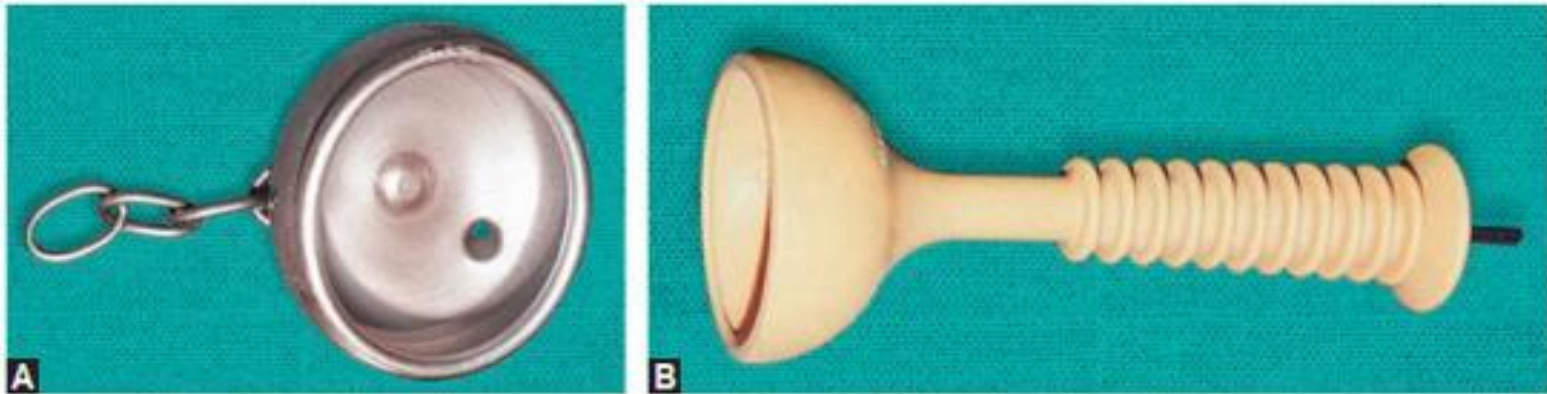
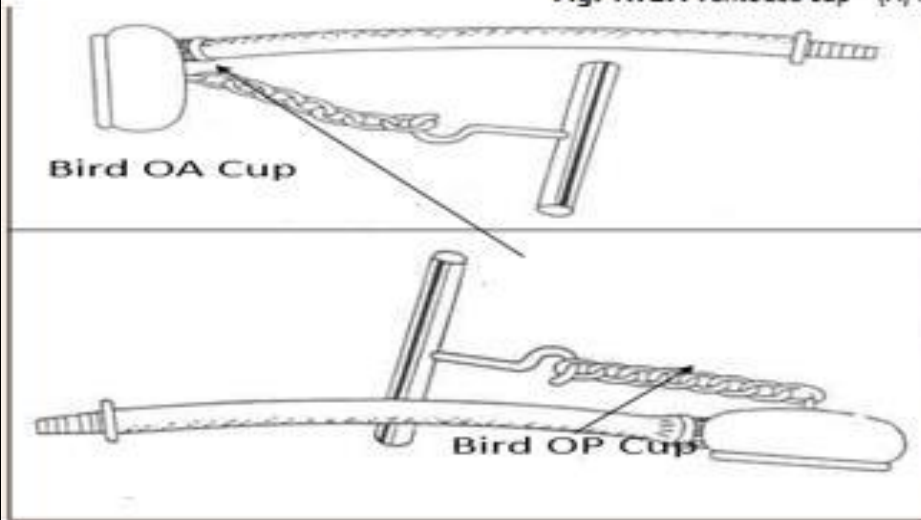


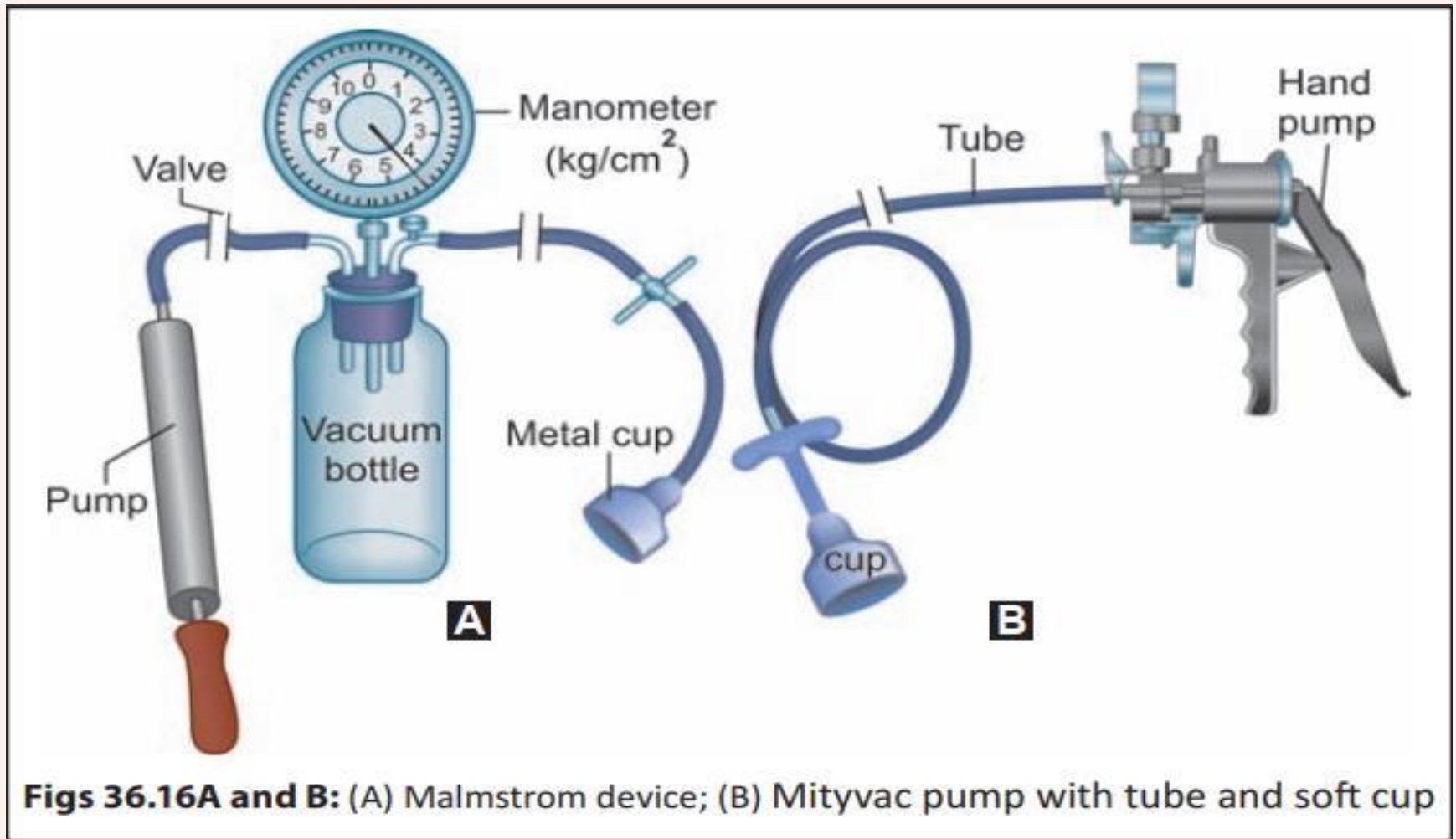
Fig. 41. 27: Ventouse cup—(A) Metal (B) Silastic



Components of vacuum

1. Cup - diameter of 3, 4, 5 or 6 cm.
2. Rubber tube -attaching the cup to a glass bottle
3. A manometer - declare the negative pressure
4. Another rubber tube - connecting the bottle to a suction piece which may be manual or electronic creating a negative pressure that should not exceed - 0.8 kg per cm².

Components of vacuum....



Indications of vacuum extraction

- Indications and pre-requisites are the same as for forceps.
- Not indicated for fetuses under 34 weeks gestational age.
- Special use-before the cervix is fully dilated ($> 8\text{cm}$)

Contraindications

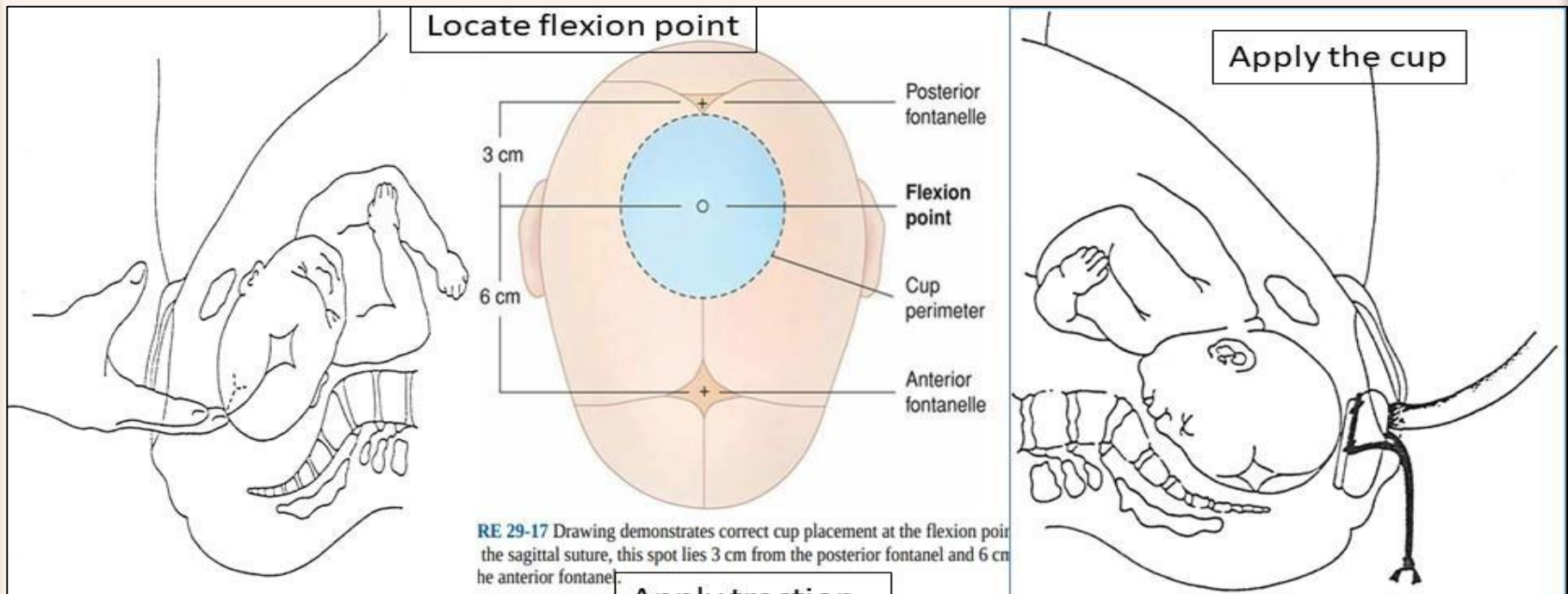
- Operator inexperience
- Inability to assess fetal position
- High station(above 0 station)
- Suspicion of cephalopelvic disproportion
- Other presentations than vertex.
- Premature fetus(<34 weeks).
- Intact membranes

Vaccume Prerequisites	• Only vertex presentation • Fully dilated cervix • Engaged head: station at 0 and below • Ruptured membranes • Gestational age 34 weeks and above • Adequate pelvis • No contraindication to vaginal delivery
Preparation	• Counsel and get written consent. • Empty bladder • Local anesthesia infiltration for episiotomy if required. • Assemble, check all connections and test the vacuum on a gloved hand

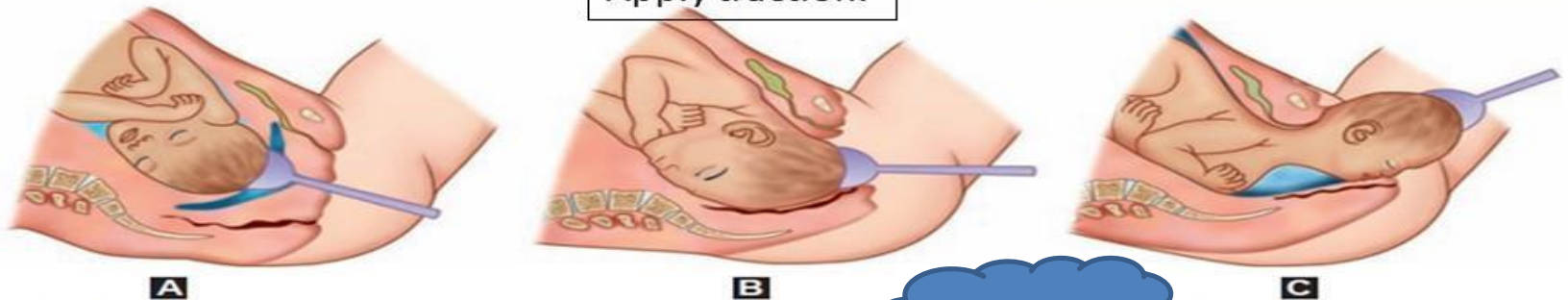
Procedure

Step	Procedure	
1	Locate flexion point	
2	Apply the cup	
3	Create a vacuum.	• Create a vacuum of 0.2 kg/cm² negative pressure and check the application of the cup. • Increase the vacuum to 0.8 kg/cm² and check the application of the cup again.
4	Apply traction.	• Applied at the onset of each contraction • First pull for right direction for descent and causes flexion. • Between contractions check: ⇒ Fetal heart rate ⇒ Application of cup ⇒ Progress of descent
5	Encourage the woman to assist descent with expulsive efforts	
6	Removal of cup.	

Procedure



Apply traction.



Figs 36.17A to C: Application of vacuum extractor; A, B, C, indicating the cup placement at different stations of the fetal head. Flexion or pivot point is an imaginary site located midsagittally about the center of the anterior fontanellae or about 2 cm in front of the posterior fontanellae. Traction over this site either by the vacuum or forceps promotes flexion and presents the smaller diameter to the pelvis (Fig. 36.16A).

Failure of vacuum

- Vacuum extraction is considered failed if:
 - ⇒ The head does not advance with each pull.
 - ⇒ The fetus is undelivered after three pulls.
 - ⇒ The fetus is not delivered within 30 minutes.
 - ⇒ The cup that is applied appropriately and pulled in the proper direction with maximum negative pressure slips off the head twice.



NB: If vacuum delivery fails, the fetus should be delivered by Cesarean section.

Advantages of vacuum over forceps

- Regional anesthesia is not required.
- Cup doesn't increase diameter beside the head unlike forceps.
- Less compression force($0.77 \text{ Vs } = 1.3 \text{ kg/cm}^2$ forceps) to head.
- Less genital tract lacerations.
- Can be applied before full cervical dilatation($\geq 8\text{cm}$)

Complications vacuum delivery

Maternal	Fetal
• Perineal, vaginal ,labial, periurethral and cervical lacerations. • Annular detachment of the cervix (incompletely dilated) • Cervical incompetence • Future prolapse if used with incompletely dilated cervix.	• Cephalohematoma. • Scalp lacerations • Subgaleal hematomas • Intracranial hemorrhage. • Neonatal jaundice • Subconjunctival hemorrhage • Retinal hemorrhage

Complications vacuum delivery.....



Figure 24: Bicornous cephalohematoma-looks like a horn

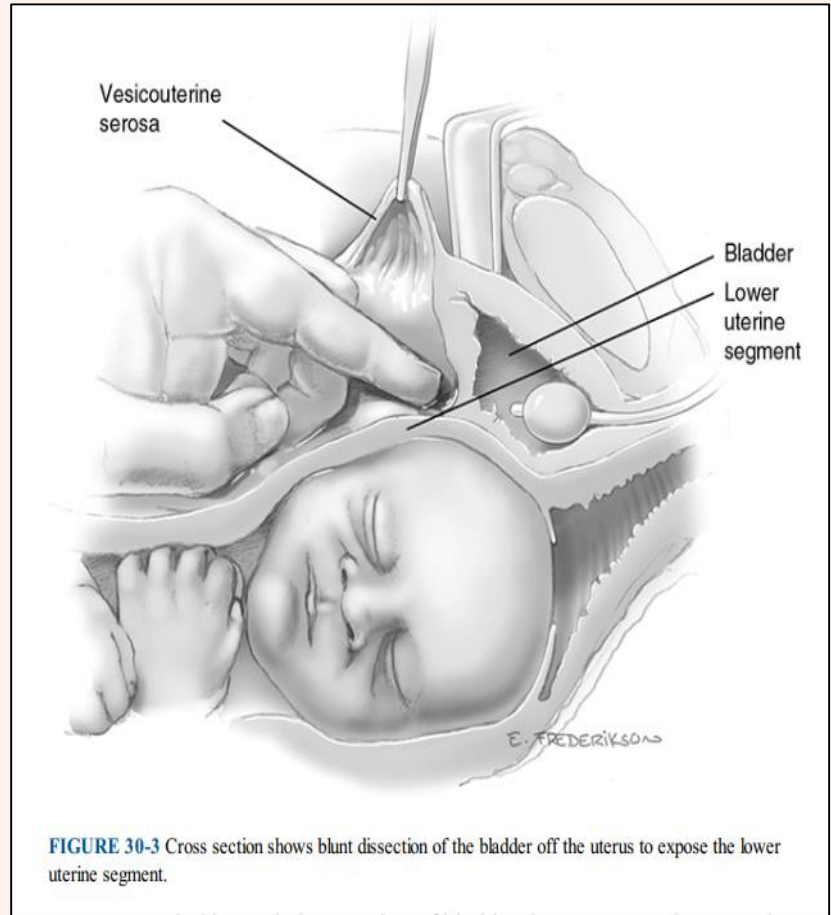


Operative vaginal delivery and PMTCT

- As much as possible operative vaginal delivery should be avoided in mother with ARVI.
- When instrumental delivery is indicated or mandatory .
 - ⇒ Limit use of vacuum extraction
 - ⇒ Prefer obstetric forceps whenever

Cesarean section(CS) /delivery

- Is the delivery of the fetus, placenta and membranes through an incision on the abdominal and uterine wall at or after 28 weeks of gestation.



Type of CS

Elective CS	• Is a planned cesarean delivery • Performed before the onset of labor or the appearance of any complication that might constitute an urgent indication.
Emergency CS	• Is when the CS is done in labor or due to any complication that necessitates immediate delivery.

Indications

ABSOLUTE INDICATIONS	RELATIVE INDICATIONS
Vaginal delivery is not possible. Cesarean is needed even with a dead fetus Indications are few: 1. Central placenta previa 2. Contracted pelvis or cephalopelvic disproportion (absolute) 3. Pelvic mass causing obstruction (cervical or broad ligament fibroid) 4. Advanced carcinoma cervix	Vaginal delivery may be possible but risks to the mother and/or to the baby are high More often multiple factors may be responsible 1. Cephalopelvic disproportion (relative) see p. 352 2. Previous cesarean delivery (p. 330)—(a) when primary CS was due to recurrent indication (contracted pelvis). (b) Previous two CS (c) Features of scar dehiscence. (d) Previous classical CS 3. Non-reassuring FHR (fetal distress) 4. Dystocia may be due to (three Ps) relatively large fetus (passenger), small pelvis (passage) or inefficient uterine contractions (Power)

Investigations

- Hemoglobin / hematocrit.
- Blood group (ABO) and Rh.
- HIV
- HBsAg
- At least 2 units of cross matched blood

Prerequisites

- Appropriate indication.
- Competent team of providers.
- Appropriate facility and equipments.

Preoperative preparation

- Elective CS
 - ⇒ Plan on days
 - ⇒ Is done at 39 weeks
 - ⇒ NPO for 8 hours for regular meal and 2 hours for clear fluid
- Emergency CS
 - ⇒ Cannot be planned
 - ⇒ Limit feeding to fluid diet in laboring women

Preoperative preparation....

- Obtain informed written consent.
- Preparation of blood (if required).
- Secure IV (16 or more gauge cannula).
- Catheterization.
- Prophylactic antibiotic (ampicillin, cephalixin, cefazolin...)
- Administer antacid solution PO (or through NG tube).
- Make sure the necessary drugs and equipment are in place.
- Make sure neonatal resuscitation set and personnel are in place.
- Check fetal heart beat before proceeding to the CS.

Intra-operative care

- Record maternal vital signs before anesthesia.
- Position on operation table(tilt to left)
- Monitoring of vital signs during the CS.
- After delivery of the baby administer 20 IU oxytocin in 1000 ml of N/S or R/L at 60 drops per-minute for two hours.

Types of skin during CS

- Types of skin during CS
1. Midline vertical
 2. Suprapubic transverse

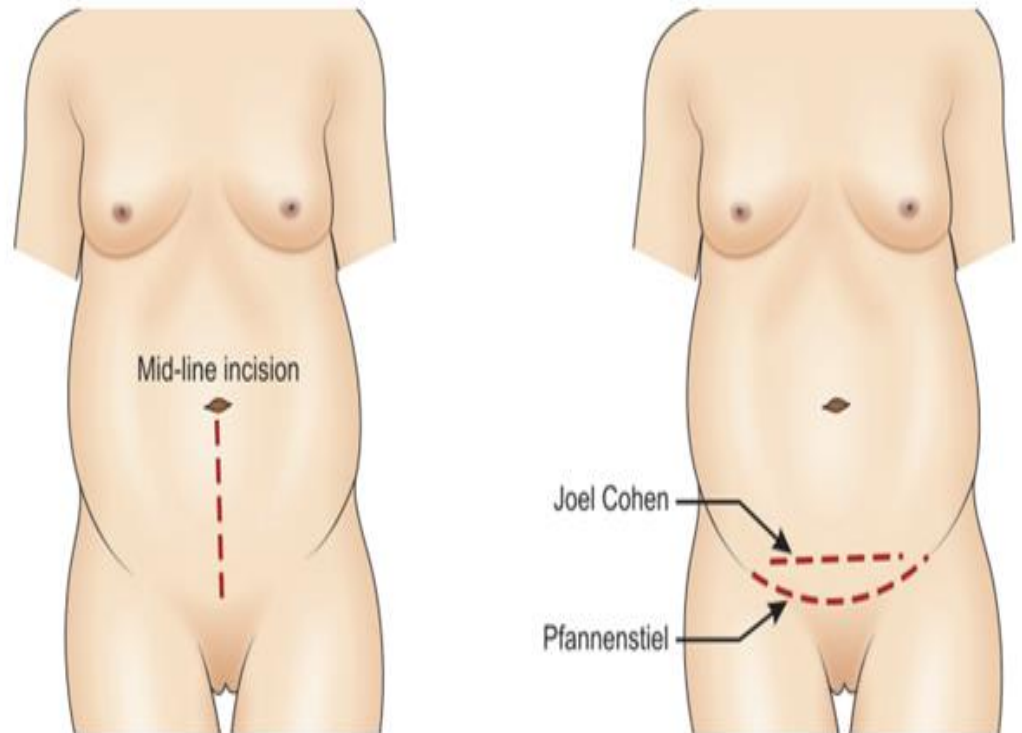


Fig. 7.2: Type of abdominal skin incision

Types of uterine incision for CS

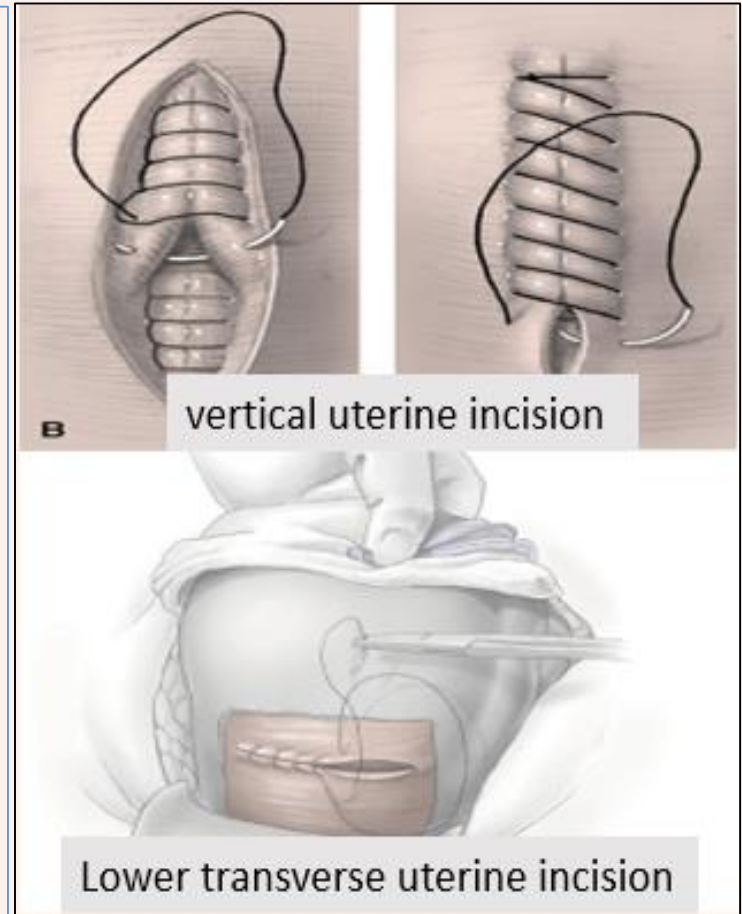
1. Lower transverse uterine incision

⇒ Commonest

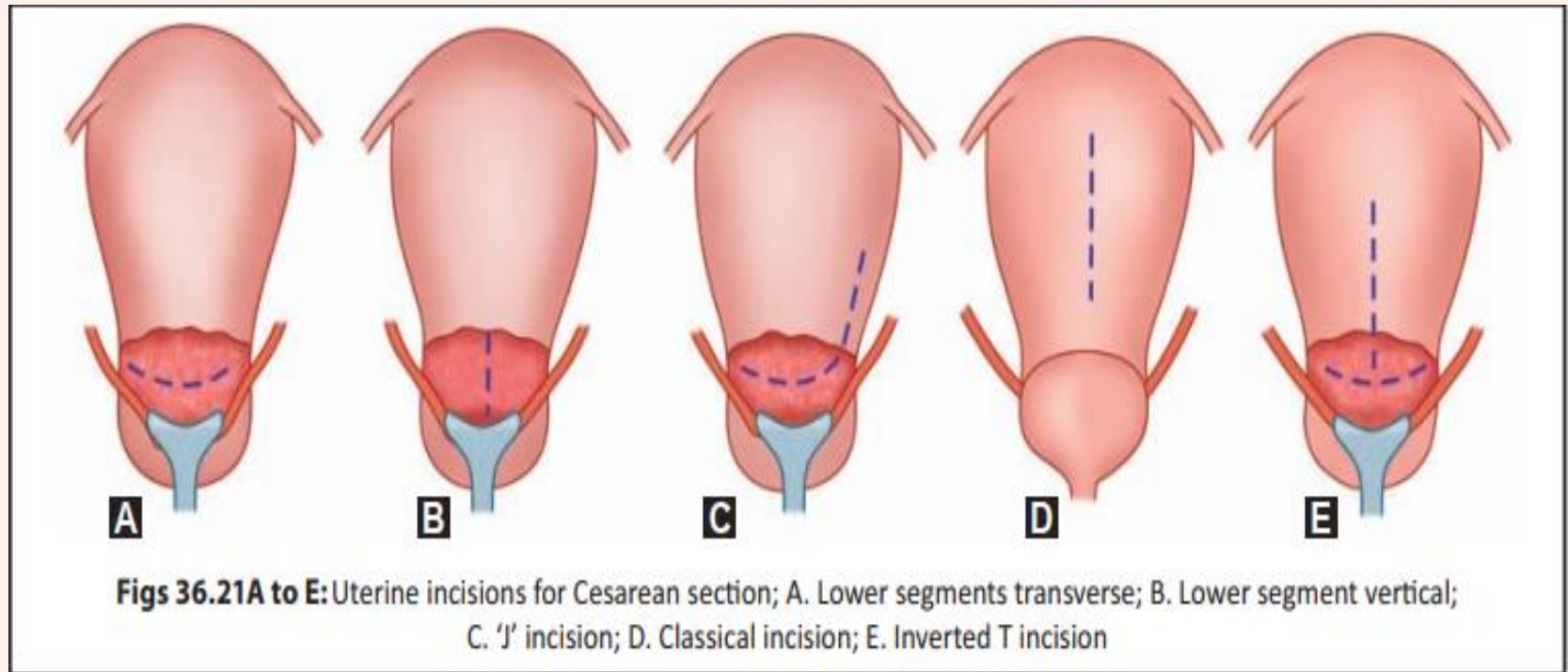
⇒ Preferable type

2. Lower vertical uterine (De Lee) incision.

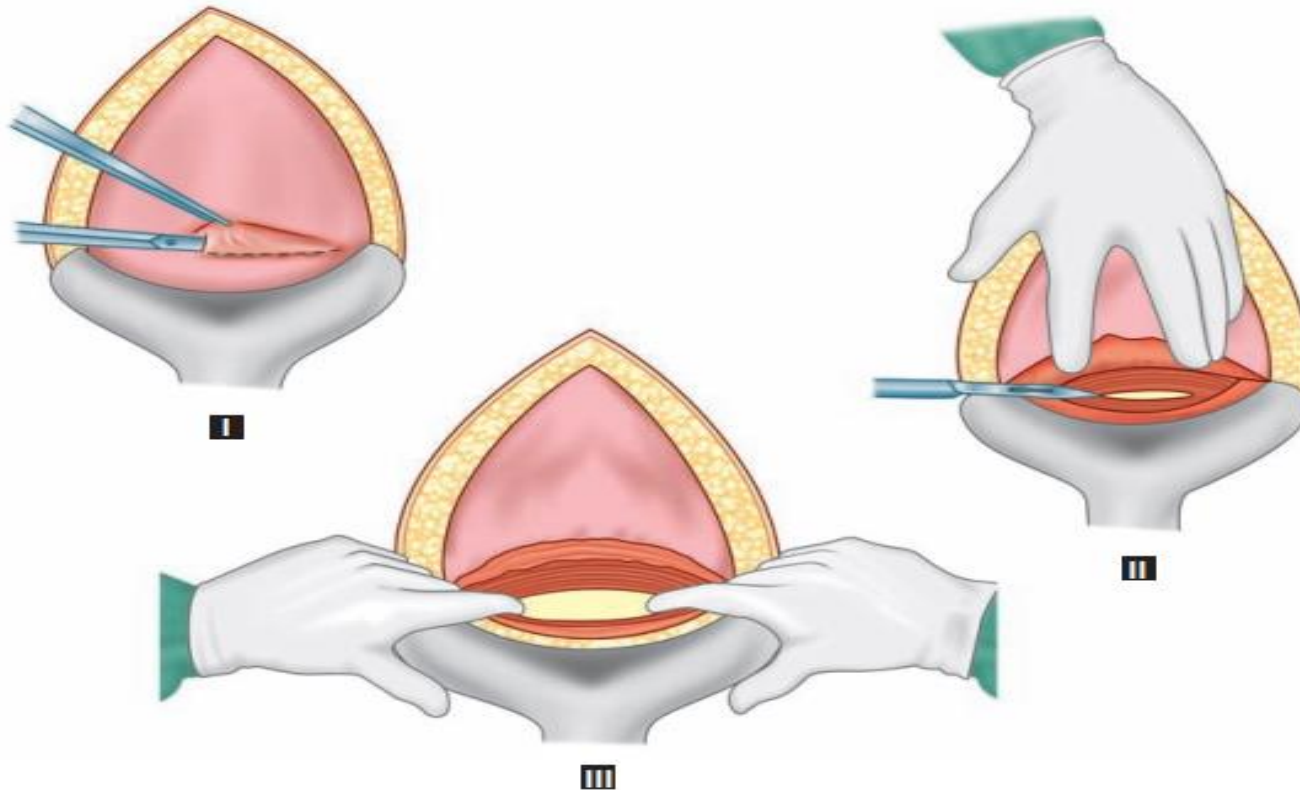
3. Upper vertical uterine incision (classical incision).



Types of uterine incision for CS...



Types of uterine incision for CS...



Figs 36.22I to III: Steps of LSCS : (I) The loose peritoneum on the lower segment is cut transversely; (II) A short incision is made in the midline down to the membranes; (III) The incision of the lower segment is being enlarged using index finger of both hands.

Steps contd.

Types of uterine incision for CS...

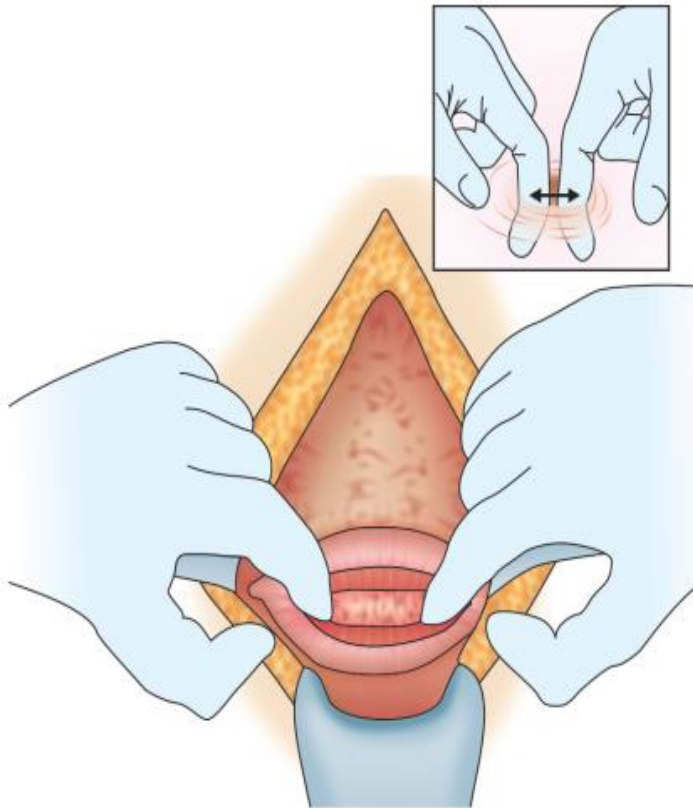


Fig. 7.3A: Extension of the uterine incision by manual stretching using the index fingers

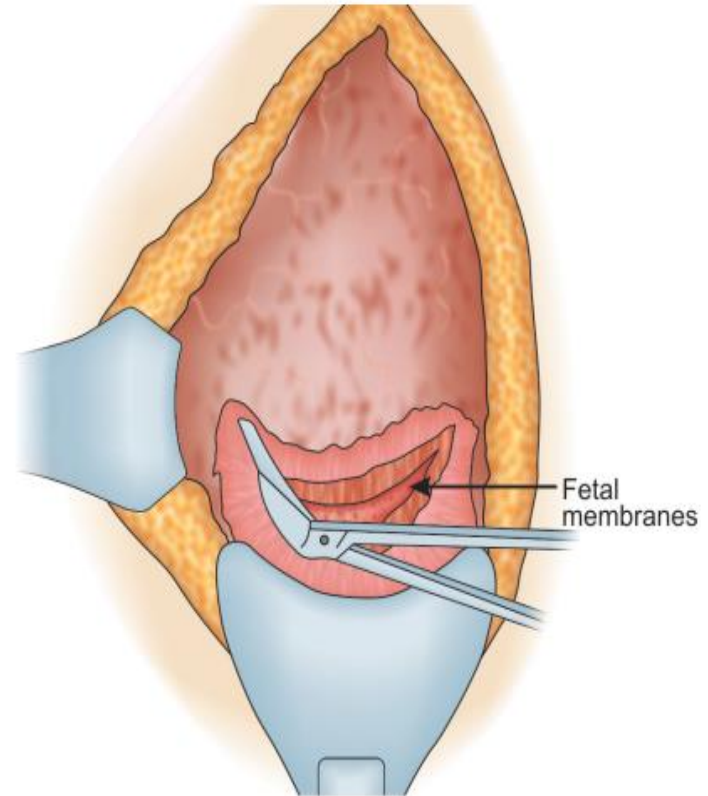
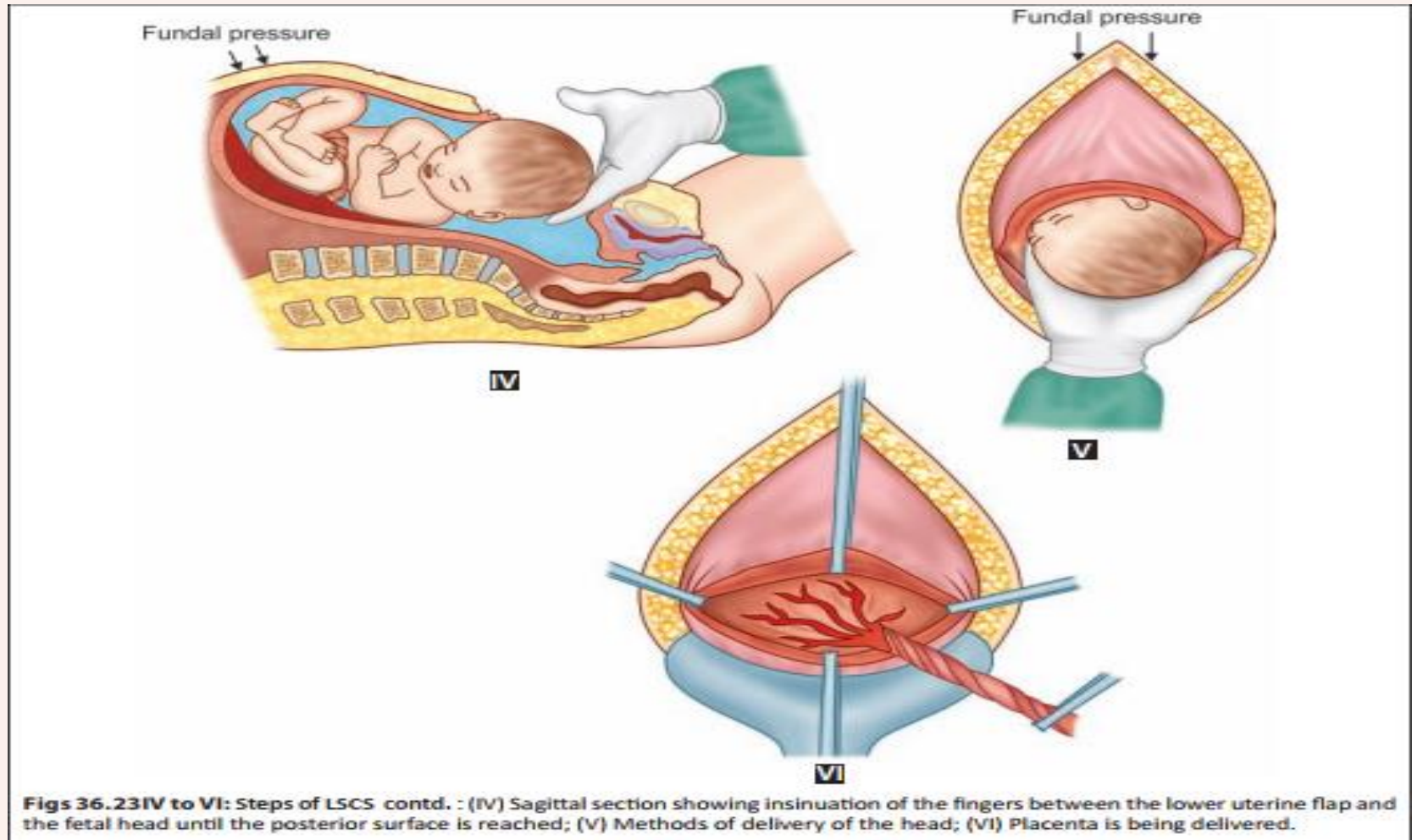
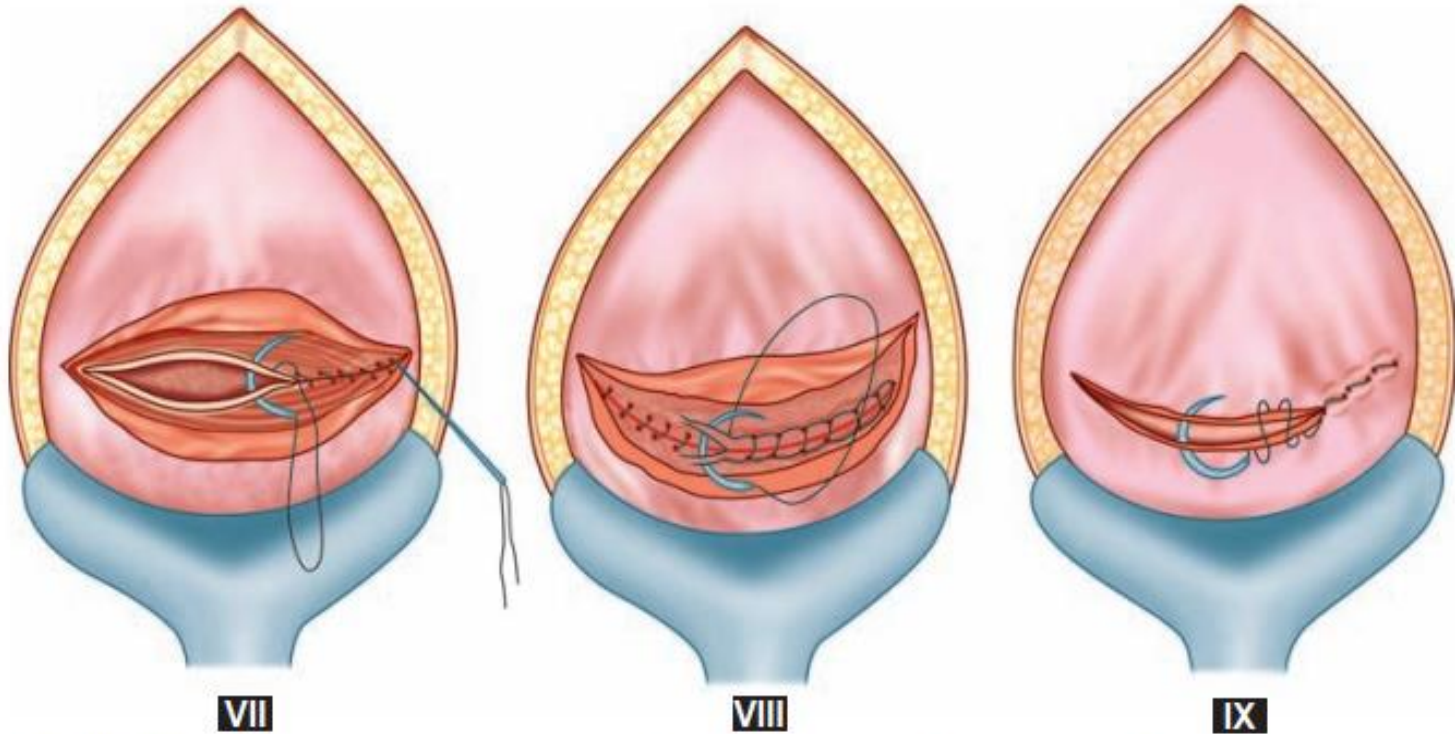


Fig. 7.3B: Extension of the uterine incision with help of scissors

Types of uterine incision for CS...



Types of uterine incision for CS...



Figs 36.24VII to IX: Steps of LSCS contd. : (VII) Inserting the continuous catgut (No. '0') suture taking deeper muscles excluding the decidua (VIII) Similar method of continuous suture taking superficial muscles and fascia down to the first layer of suture (IX) Continuous peritoneal catgut suture.

Types of uterine incision for CS

	<i>Lower segment transverse</i>	<i>Lower segment vertical</i>
♦ Extension of incision	- may occur to involve the uterine vessels	- may occur to involve the upper segment or downward the bladder
♦ Bladder dissection	- minimal	- more when extends inferiorly
♦ Uterine closure	- easy	- difficult
♦ Muscle apposition	- good	- often imperfect
♦ Reperitonization	- complete	- often imperfect
♦ Intraoperative bleeding	- less	- more
♦ Subsequent adhesions	- less	- more

Postoperative care

- Vital signs every 15 min until awake, then every hour for 4 hours
- Check vaginal bleeding and uterus is contracted.
- **Check** and record urine output every 4 - 6 hours.
- Provide analgesics as required.
- Initiate breast-feeding soon as the mother is awake.
- Start sips of fluid (conscious and bowel sounds)
- Discontinue IV fluids once fluid diet is started
- Ambulate early

Unit 6: Abnormal puerperium

1

- Post partum hemorrhage

2

- Puerperal fever

3

- Lactational mastitis

Abnormal puerperium...

- Objective
 - ⇒ Define PPH, cause presentation and management
 - ⇒ Describe risk ,cause presentation and management of puerperal fever
 - ⇒ Discuss on lactational mastitis(risk, cause, presentation and management)

Postpartum hemorrhage

1. A blood loss that exceeds

⇒ 500 ml after a vaginal birth

⇒ 1,000ml after a cesarean birth

⇒ 1500 ml after hysterectomy

2. Hematocrit Change – defined as change $> 10\%$

3. Hemodynamic instability(vital sign derangement)

4. Need for transfusion – variable practice

Classifications

Primary, early or acute PPH	Secondary or Late PPH
• Delivery - < 24hours PP • 90% PPH cases • Associated with more bleeding	• Affects 1-3 % of all deliveries • Common causes in the first 3 weeks include: ⇒ Sub-involution of the uterus ⇒ Intrauterine Infection (e.g Endometritis) ⇒ Retained pieces of placental tissue(RPOC)

Causes PPH

Early	Late
• Uterine atony • Lower genital tract lacerations • Upper genital tract lacerations • Lower urinary tract lacerations • Retained products of conception • Invasive placentation • Uterine rupture • Uterine inversion • Coagulopathy	• Infection • Retained products of conception • Placental site subinvolution • Coagulopathy

Uterine atony

- Also called uterine inertia
- Inability of the uterine myometrium to contract effectively.
- The most common cause of early postpartum hemorrhage
- The most typical causes in the **1st 4 hours**
- **75-90%** PPH
- Bleeding is from placental implantation site

Risk factor of uterine atony

- Uterine over distention
 - ⇒ Multiple pregnancy
 - ⇒ Macrosomia
 - ⇒ Polyhydramnios
- Weakness of uterine muscle
 - ⇒ Prolonged Labor
 - ⇒ Rapid Labor
 - ⇒ Induced/augmented labor
 - ⇒ Grandmaltiparity
 - ⇒ Advanced age(>35)
 - ⇒ Intrinsic myometrial dysfunction
 - ⇒ General anesthetics- Halothane
- Functional/anatomic distortion of the uterus
 - ⇒ Fibroid uterus(myoma)
 - ⇒ Abnormal placentation
 - ⇒ Uterine anomalies
- Poorly perfused myometrium-hypotension
 - ⇒ Hemorrhage, conduction analgesia
- Couvelaire uterus
- Previous uterine atony

Diagnosis of atony

- Bimanual palpation of the uterus confirms the diagnosis.
 - ⇒ Soft or “ boggy” feel when fundus is located
 - ⇒ Uterine fundus that is located above expected level
 - ⇒ Uterus that becomes firm when massaged but loses its tone
 - ⇒ Uterus not well contracted
 - ⇒ Soft & Lax uterus

Uterine atony management

- Shout for help
- Rapid initial assessment(ABC)
- Left lateral position
- Treat shock if she develop
- Massage uterus to expel blood and tissues
- Give 10 IU oxytocin
- Take 5ml blood for laboratoy
- Catheterization
- Examine the placenta and the membrane
- Treat accordingly the cause

Uterine atony management....

- Fluid replacment
 - ⇒ Use large bare cannula bilaterally
 - ⇒ 1st 1000ml rapidly in 15-20 minute
 - ⇒ At least 2000ml in the 1st hour.
 - ⇒ After she is stable,adjust rate 1000ml/6 hour.
- Aim⇒ Replace blood loss three times

Uterine atony management....

- Massage uterus
- Use uterotonic agent
- Empty bladder and breast feeding

Table 18-6. Uterotonic Therapy

AGENT	DOSE	ROUTE	DOSING FREQUENCY	SIDE EFFECTS	CONTRAINDICATIONS
Oxytocin (Pitocin)	10–80 units in 1,000 ml of crystalloid solution	First line: IV Second line: IM or IU	Continuous	Nausea, emesis, water intoxication	None
Methylergonovine (Methergine)	0.2 mg	First line: IM Second line: IU or PO	Every 2–4 h	Hypertension, hypotension, nausea, emesis	Hypertension, preeclampsia
15-Methyl Prostaglandin F _{2α} (Hemabate)	0.25 mg	First line: IM Second line: IU	Every 15–90 min (8-dose maximum)	Nausea, emesis, diarrhea, flushing, chills	Active cardiac, pulmonary, renal or hepatic disease
Prostaglandin E ₂ (Dinoprostone)	20 mg	PR	Every 2 h	Nausea, emesis, diarrhea, fever, chills, headache	Hypotension
Misoprostol (Cytotec)	600– 1000 mcg	First line: PR Second line: PO	Single dose	Tachycardia, fever	None

Conservative management of uterine atony

- If pharmacologic methods fail to use
 - ⇒ Bimanual compression of uterus
 - ⇒ Aortic compression if bleeding continue
 - ⇒ Uterine tamponade
 - ⇒ Selective arterial embolization
 - ⇒ Compression sutures.
 - ⇒ Vessel ligation.
 - ⇒ Surgical intervention.

Uterine atony prevention

1. Active management of the third stage of labor(AMTSL)
2. Spontaneous delivery of the placenta after cesarean delivery
 - Oxytocin
 - Controlled cord traction
 - Massage uterus
3. Avoid risk factor

Trauma

- Is the second most frequent causes 20%
- Clinical finding
 - ⇒ Firm & well contracted Uterus
 - ⇒ Bright red persistent bleeding,oozing,or frank hemorrhage
- Careful inspection of vagina,cervix, perineum
- Treated with repaired

Retained placenta

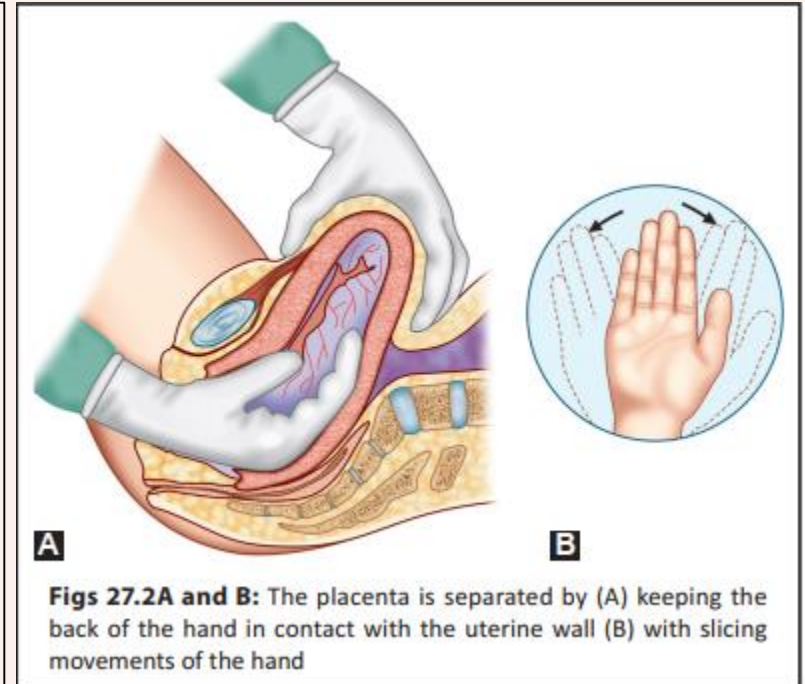
- Cause 5–10% of postpartum hemorrhages.
- Placenta is not expelled 30 minutes after delivery
- Placental tissue and amniotic membranes inhibit uterus from adequate contraction and result in hemorrhage.
- 10% placenta's have fundal implantation

Retained placenta...

- Possible cause
 - ⇒ Mismanagement of the third stage of labor
 - ⇒ Partial separation of a normal placenta
 - ⇒ Cervical constriction
 - ⇒ Failure of the woman to push out
 - ⇒ Closure of the cervix
 - ⇒ Uterine inertia
 - ⇒ Pathological adherence
- Manage with manual separation and removal

Retained placenta management

- Ensure bladder is empty
- Manual removal of placenta
- Pethidine and diazepam
- Ampicillin 2g + Metronidazole 500mg
- Check completeness
- Oxytocin 20 U/1000 ml 60dro/min
- Monitor vital sign



Late postpartum hemorrhage

- Blood loss of more than 500 ml later than 24 hrs after delivery.
- Causes :
 - ⇒ Sub-involution
 - ⇒ Retained placental fragments
 - ⇒ Infection
- Management :
 - ⇒ Uterotonic drug administration
 - ⇒ Curettage
 - ⇒ Broad spectrum antibiotics if there are signs of infection

Puerperal fever

- Also known as postpartum fever is defined as temperature of 38.0°C or higher during the first 10 days postpartum, exclusive of the first 24 hours.
- Fever in the first 24 hours after delivery
 - ⇒ Often resolves spontaneously
 - ⇒ Cannot be explained by an identifiable infection.

Cause of puerperal fever

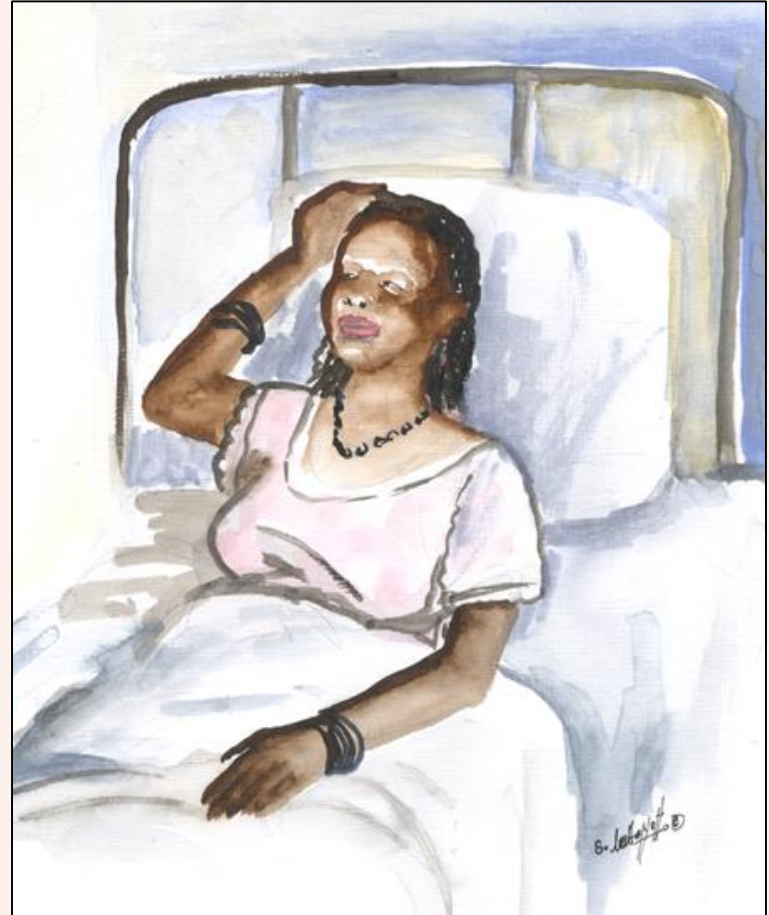
- Endometritis
- Urinary tract infection
- Mastitis
- Genital tract infection.
- Breast engorgement
- Wound infections(episiotomy,incisions or perineal lacerations)
- Respiratory complications after caesarean delivery
- Septic thrombophlebitis.
- Endemic febrile illnesses like malaria

Puerperal fever...

- The most common cause of postpartum fever is endometritis, which occurs after vaginal delivery in about 2% of patients and after cesarean delivery in about 10 % to 15%.

Endomyometritis

- Is inflammation of the endometrium the inner lining of the uterus and myometrium
- The most cause puerperal pyrexia
- Is a major cause of maternal death



Risk factors

- Prolonged and premature ROM
- Prolonged (> 24 hours) labor
- Frequent vaginal examination
- Retained placental fragments or membranes
- Anemia and poor nutrition during pregnancy
- Immunocompromised
- Genital or urinary tract infection prior to delivery
- Cesarean birth (20-fold increased risk)
- Obesity
- Diabetes
- Indwelling urinary catheter.

Cause of endometritis

- The vaginal flora
 - ⇒ Doderlein's bacillus (60–70%)
 - ⇒ Candida albicans (25%)
 - ⇒ Staphylococcus albus or aureus
 - ⇒ Streptococcus—anaerobic common; beta hemolyticus rare
 - ⇒ E. coli and Bacteroides group

Clinical features

- Fever
- Lower abdominal pain
- Offensive lochia
- Subinvoluted uterus
- Lower abdominal tenderness
- Uterine tenderness.
- Less common presentation
 - ⇒ Abdominal distension, nausea and vomiting, diarrhoea, abdominal fluid collection, generalized abdominal tenderness and guarding, adnexal mass

Investigation

- Blood film
- CBC
- C reactive protein
- Urinalysis
- Stool exam
- Abdominopelvic ultrasound

Management

- Supportive care
 - ⇒ Encourage rest.
 - ⇒ Ensure adequate hydration by mouth or IV.
 - ⇒ Use a fan or tepid sponge
 - ⇒ Consider paracetamol 500–1000 mg QID
 - ⇒ Transfusion of blood

Management

- Antibiotics

⇒ Clindamycin phosphate 600 mg IV every eight hours PLUS Gentamicin 5 mg/kg body weight IV every 24 hours.

⇒ Ampicillin 2 g IV every 6 hours PLUS gentamicin 5 mg/kg body weight IV every 24 hours.

⇒ Give 24–48 hours after resolution of clinical signs and symptoms (fever, uterine tenderness, purulent lochia, and leukocytosis).

Prevention

- Avoid risk factors.
- Keep the episiotomy site clean.
- Careful attention to antiseptic procedures during childbirth.
- Administration of prophylactic antibiotics (e.g. cesarean section, manual removal of placenta).

Complication

- Pelvic abscess
- Peritonitis
- Septic shock
- Deep vein thrombosis
- Pulmonary embolism
- Chronic pelvic infection with recurrent pelvic pain and dyspareunia
- Tubal blockage, or infertility

Lactational acute mastitis

- Is inflammation of mastitis
- Usually peripherally situated.
- most frequently in the first 2 to 4 weeks postpartum



Risk of mastitis

- Partial blockage of milk duct
- Pressure on the breast (eg, tight brassiere or car seatbelt)
- Oversupply of milk
- Infrequent feedings
- Nipple excoriation or cracking
- Rapid weaning
- Illness in mother or baby
- Maternal stress or excessive fatigue
- Maternal malnutrition

Etiology

- The most common organisms associated with mastitis are
 - ⇒ **Staphylococcus aureus**
 - ⇒ Methicillin-resistant *S. aureus* (MRSA)
 - ⇒ *S. epidermidis*
 - ⇒ Streptococci and
 - ⇒ Occasionally, gram-negative rods.
- The incidence of sporadic mastitis is 2% to 10% in lactating mothers and less than 1% in nonlactating mothers

Clinical presentation

- Fever $>38.3^{\circ}\text{C}$
- Severe, throbbing type of pain
- Hard, red, tender, swollen area of one breast
- Myalgia, chills, malaise
- Advanced infection
 - ⇒ Large area of breast swelling (erythema)
 - ⇒ Reactive auxiliary lymphadenopathy

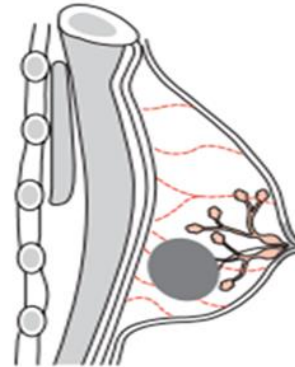


Figure 50.13 Intramammary breast abscess.

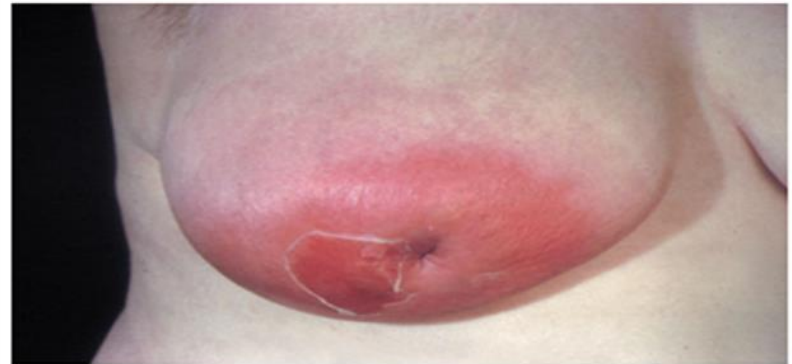


Figure 50.14 Large breast abscess.

Treatment of mastitis

- Fluids
- Encourage the woman to continue breastfeeding,
- Support the breasts with a binder or bra, and
- Apply cold compresses to the breasts between feedings
- Cloxacillin 500 mg PO every six hours for 10 days OR Erythromycin 250 mg every eight hours for 10 days
- Paracetamol 500–1000 mg every six to eight hours as an appropriate alternative (maximum dose 4000 mg in 24 hours).

Complication of acute mastitis

- Abscess
- Skin necrosis
- Toxemia
- Antibiotoma
- Fistula

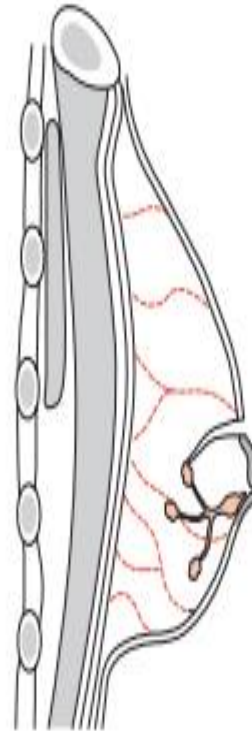


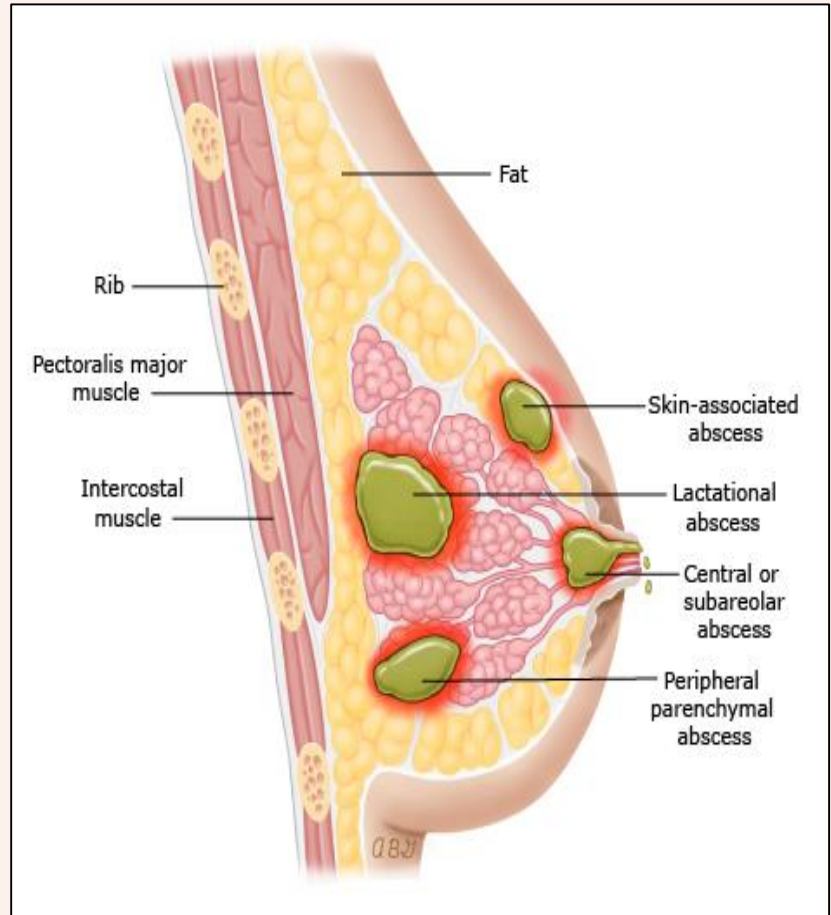
Figure 50.18 Mammary fistula originating in a chronic subareolar abscess.

Breast abscess

- Will occur in about 10% of women who are treated for bacterial mastitis.
- The signs include
 - ⇒ High fever (39° to 40° C)
 - ⇒ Unilateral breast swelling and pain
 - ⇒ Localized and fluctuant mass with erythema of overlying skin
 - ⇒ The patient feels sick, like having the “flu.”
- Abscesses usually occur in the upper outer quadrants

Management of breast abscess

- Cloxacillin 500 mg PO every six hours for 10 days OR
- Erythromycin 250 mg every eight hours for 10 days.
- Drainage of the abscess is indicated
- Breastfeeding should be limited to the uninvolved side during the initial therapy



Unit 7: Early pregnancy abnormalities

1

- Ectopic pregnancy

2

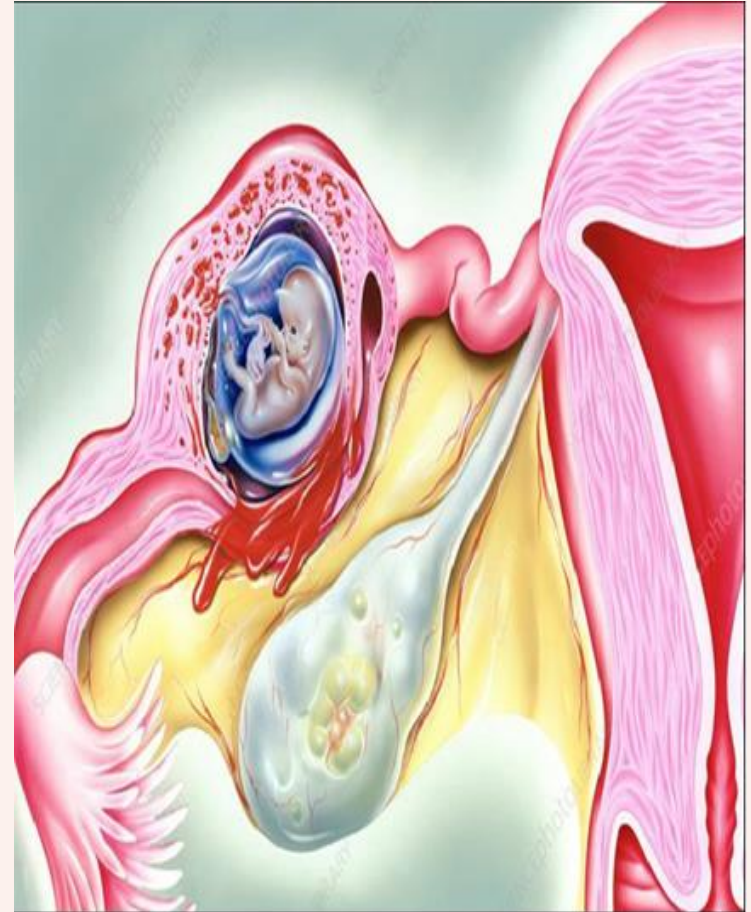
- Abortion

Abnormalities of early pregnancy

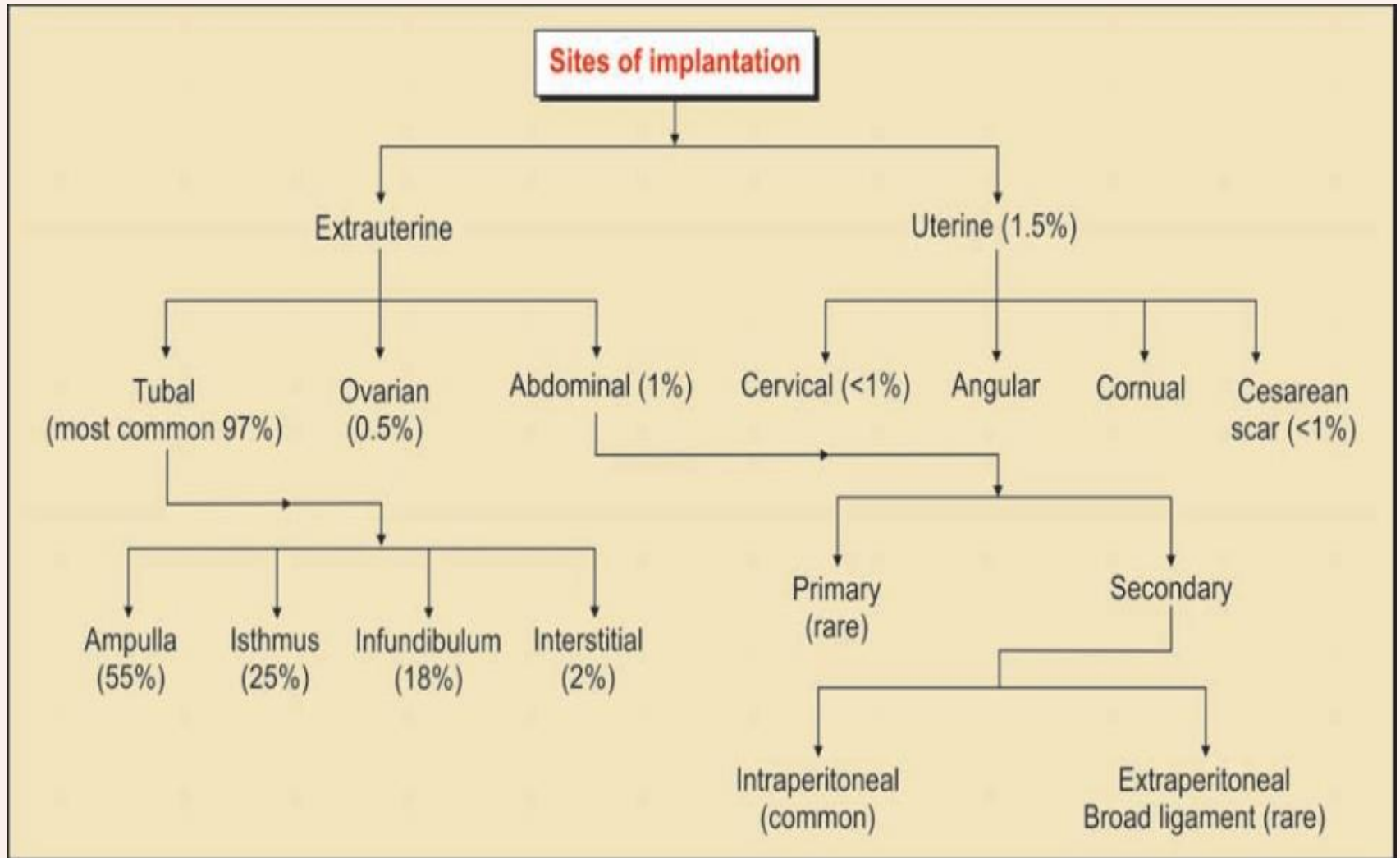
- Objective
 - ⇒ Explain risk factor, presentation and management of ectopic pregnancy
 - ⇒ Define abortion, cause ,classification and management

Ectopic pregnancy

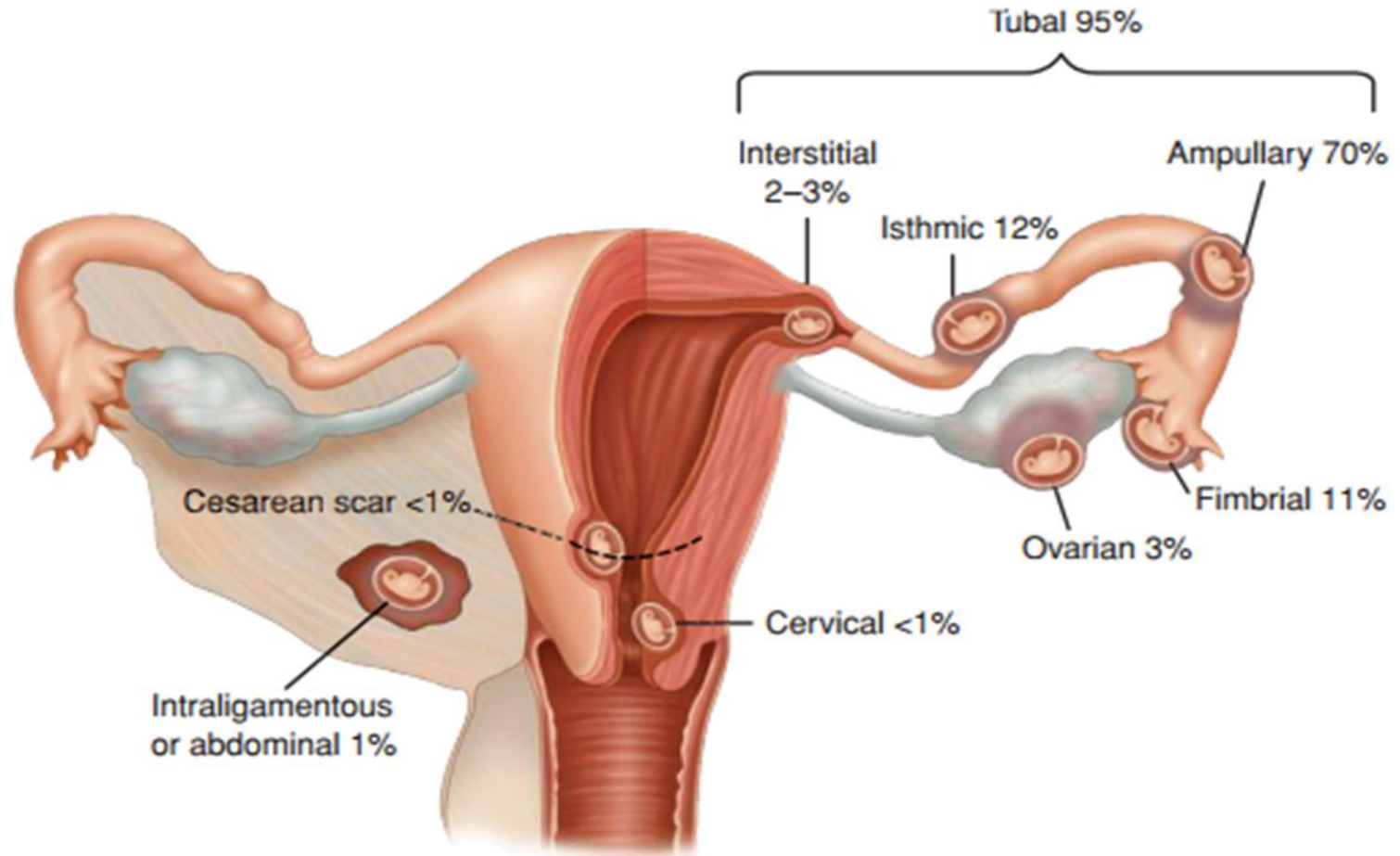
- Is implantation of a fertilized ovum outside uterine cavity.
- Classification
 - ⇒ Fallopian tube
 - ⇒ Extra-tubal
- 2% of pregnancies are ectopic
- 6% of all pregnancy-related deaths.



Site of ectopic pregnancy



Ectopic pregnancy...

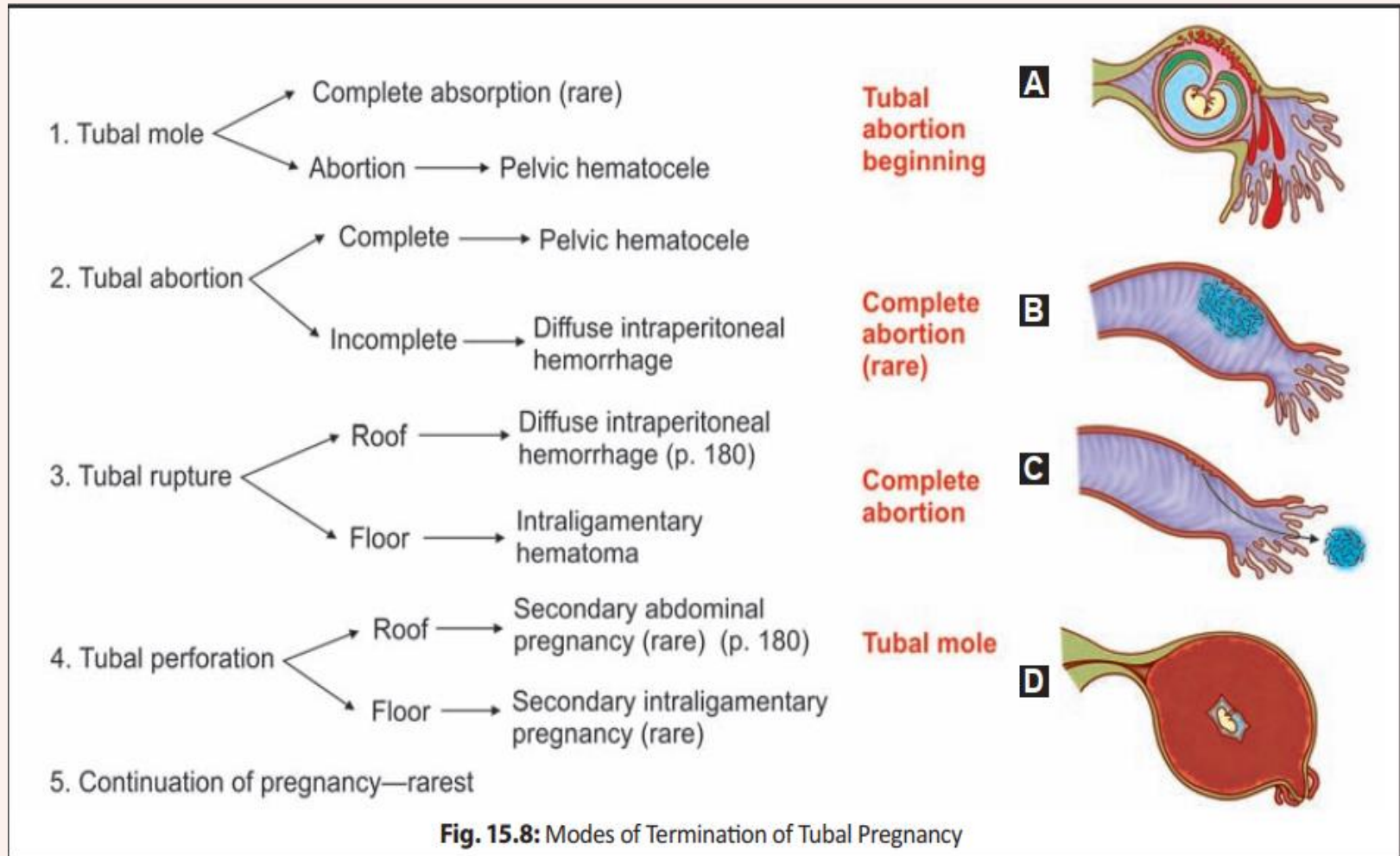


19-1 Sites of implantation of 1800 ectopic pregnancies from a 10-year population-based study. (Data from Callen, 2003)

Risk factors

- History of PID (e.g. secondary to STI)
- History of ectopic pregnancy or tubal surgery.
- History of infertility
- Contraceptive failure.
- History of tubal ligation
- IUD use
- Previous induced abortion

Natural history



Natural history....

- Tubal rupture

- ⇒ Isthmic pregnancies rupture at 6 to 8 weeks
- ⇒ Ampullary pregnancies rupture at 8–12 weeks.
- ⇒ Interstitial pregnancies rupture at 12–16 weeks

Clinical features

- Classical triads: present in 50-60%
 1. Abdominal pain: 98%
 2. Missed period: 75%
 3. Vaginal bleeding: 70%
- Early pregnancy symptoms, fainting or syncope
- Features of anemia (e.g. dizziness)
- Pain on defecation and shoulder tip pain

Signs of unruptured and ruptured ectopic pregnancies.

Sign	Unruptured	Ruptured
Vital Signs	• Vital signs can be normal or slightly deranged	• Fast, weak pulse • Hypotension • Pallor
Abdominal examination	• Non-tender abdomen	• Direct abdominal tenderness • Rebound tenderness • Abdominal distension • Bowel sounds are decreased
Pelvic examination	• Minimal vaginal bleeding • Adnexal mass	• Minimal vaginal bleeding • Cervical motion tenderness • Adnexal tenderness

Differential diagnosis

- Abortion
- PID
- Ovarian torsion
- Ruptured ovarian cyst
- Acute appendicitis



Diagnosis

- History and physical findings
- Transvaginal sonography (TVS)
- Serum β -hcG level measurement
- Hematology: CBC, Hgb, Blood group & Rh status, ESR
- Diagnostic surgery
 - ⇒ Uterine curettage
 - ⇒ Laparoscopy
 - ⇒ Laparotomy

Management

- Secure IV line with large bore canula.
- Cross-match blood and prepare for immediate laparotomy.
- Do not wait for blood before performing surgery.
- Surgery
 - ⇒ Salpingectomy
 - ⇒ Salpingostomy
 - ⇒ Salpingotomy

Subsequent management

- Correct anemia
- Anti-D should be administered to all Rh-negatives
- Provide family planning counseling and service.
- Counseling and advice on prognosis of fertility
- Counseled to come as soon as she misses her menses.
- Schedule a follow-up visit.

Abortion

- Is defined as the spontaneous or induced termination of pregnancy before fetal viability.
- ⇒ WHO define abortion as pregnancy termination before 20weeks of gestation or with fetus born weighing < 500g.
- ⇒ Ethiopia and UK the age of viability is 28weeks or <1000gm.
- It is the most common complication of pregnancy

Incidence

- Is difficult to work out
- 10–20% of all clinical pregnancies end in miscarriage
- About 75% miscarriages occur before the 16th week
 - ⇒ Of these 80% occur before 12th week of pregnancy.

Etiology

First trimester	Second trimester
1. Genetic factors (50%) 2. Endocrine disorders • Luteal phase defect • Thyroid abnormalities • Diabetes 3. Immunological disorders 4. Infection 5. Unexplained.	1. Anatomic abnormalities • Cervical incompetence • Mullerian Fusion defects ⇒ Bicornuate uterus ⇒ Septate uterus • Uterine synechiae • Uterine fibroid 2. Maternal medical illness 3. Unexplained

Classification of abortion

By occurrence	A. Spontaneous(miscarriage)	
	B. Induced abortion	1. Therapeutic abortion 2. Elective(voluntary) abortion
Based on GA	1. 1st TM vs 2nd TM 2. Early(<12wks) vs Late(12-28wks)	
Based on place of termination	1. Safe 2. Unsafe	
Clinical	1. Threatened 2. Inevitable 3. Incomplete 4. Complete 5. Missed abortion 6. Septic abortion 7. Recurrent/habitual abortion	

Threatened abortion

- The process of miscarriage has started but has not progressed to a state from which recovery is impossible
- Clinical presentation
 - ⇒ Mild vaginal bleeding
 - ⇒ No or mild pain usually following hemorrhage.
 - ⇒ Closed cervix or intact membrane

Threatened abortion...

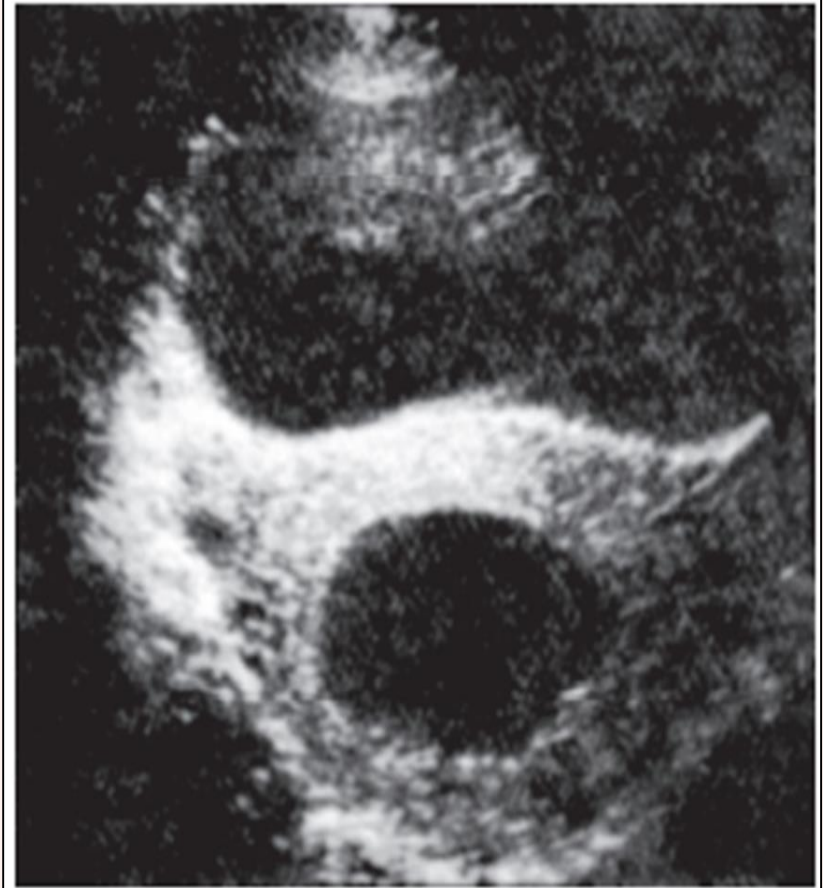
- Investigation

- ⇒ CBC,BG &RH

- ⇒ Urine

- ⇒ hcG test

- ⇒ Confirmed by ultrasound which shows a blighted ovum



Management

- Reassurance (50% probability of continue)
- Decrease physical activity (bed rest), avoid intercourse
- Hormones like Progesterone & hCG
- Treat complication
- Anti- D
- Council on outcome
 - ⇒ Antepartum hemorrhage
 - ⇒ 50 will progress to incomplete abortion
 - ⇒ Are at increased risk of preterm labor

Inevitable abortion

- Continuation of pregnancy is impossible
- Clinical features
 - ⇒ Increased vaginal bleeding.
 - ⇒ Mild to moderate lower abdomen pain
 - ⇒ No passage of products conception
 - ⇒ Dilated internal os with protruded conceptus tissue
 - ⇒ Uterus normal for date
 - ⇒ Leakage of liquor even without open cervix
 - ⇒ The diagnosis is made by clinically.

Management

- CBC , blood grouping , cross match 2 units of blood
- Resuscitation
- AntiD
- Oxytoxic drugs
- Evacuation & curettage.
- Post-abortion management.

Complete abortion

- All products of conception are expelled
- Clinical features
 - ⇒ History of expulsion of a fleshy mass per vagina
 - ⇒ Subsidence of abdominal pain.
 - ⇒ Vaginal bleeding becomes trace or absent.
 - ⇒ Uterus is smaller than the period of amenorrhea
 - ⇒ Cervical os is closed
 - ⇒ Ultrasonography (TVS): reveals empty uterine cavity.

Management

- Post-abortion management
- Ways of confirming completeness
 1. Examine concepts carefully for completeness
 2. Ultrasound to see retained tissue
 3. Documented completeness on referral paper
- If any doubt of completeness evacuate the uterus
- Administer ergometine 0.5mg if bleeding

Incomplete abortion

- Some part of products of conception is expelled and other is not .
- The commonest type
- Clinical features
 - ⇒ Sever lower abdomen.
 - ⇒ Persistence of vaginal bleeding with expulsion of tissue
 - ⇒ Uterus smaller than the period of amenorrhea
 - ⇒ Dilated cervical os often admitting tip of the finger

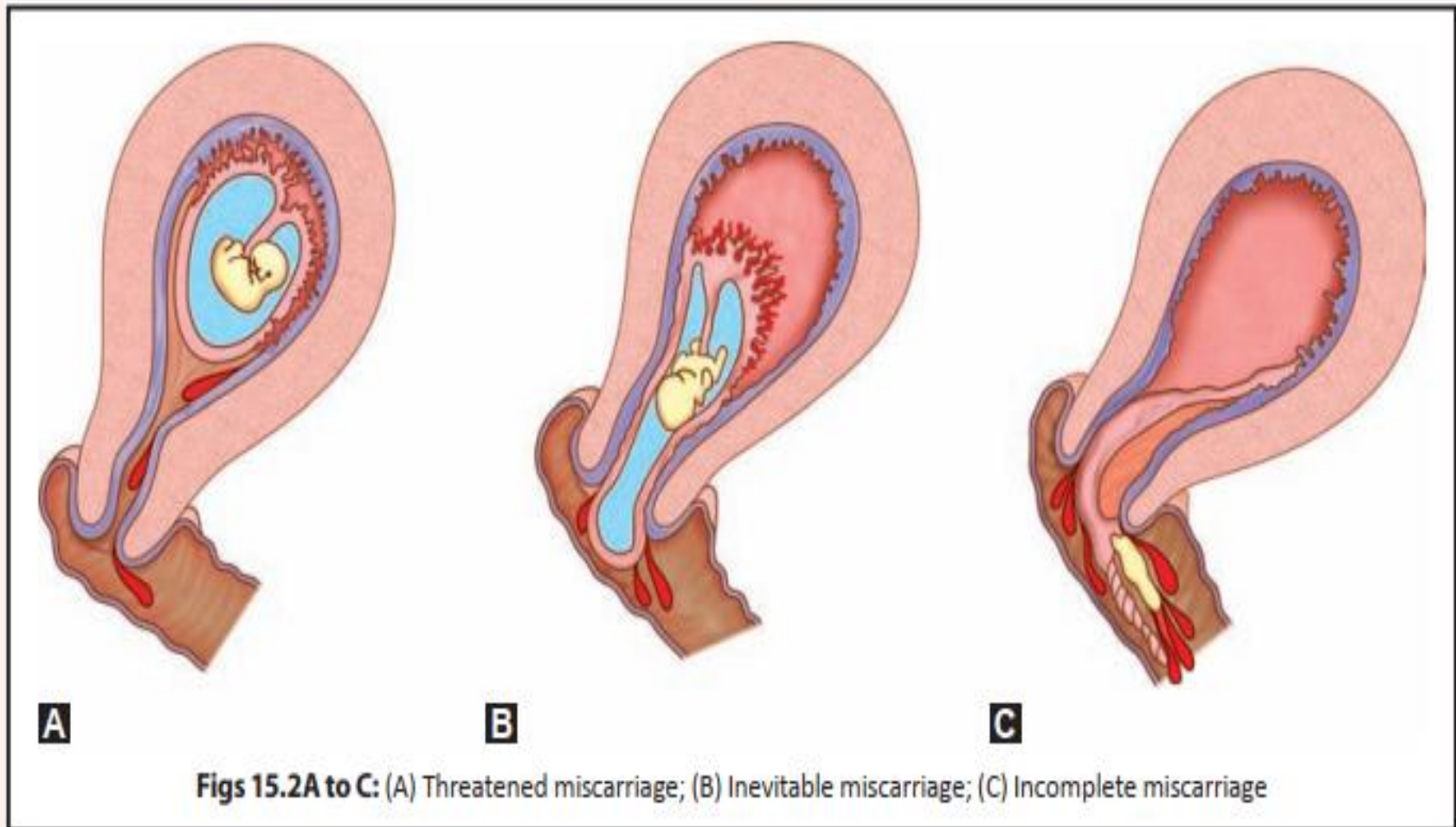
Incomplete abortion...

- Ultrasonography reveals echogenic material in uterus.
- Complications
 - ⇒ Profuse bleeding
 - ⇒ Sepsis
 - ⇒ Polyp

Management

- CBC , blood grouping , cross match 2 units of blood
- Resuscitation
- AntiD
- Uterotonic drugs
- Evacuation & curettage.
- Post-abortion management.

Classification of abortion...



Missed abortion

- Fetus is dead and retained inside uterus for a variable period
- Clinical features
 - ⇒ Persistence of brownish vaginal discharge.
 - ⇒ Subsidence of pregnancy symptoms
 - ⇒ Cessation of uterine growth
 - ⇒ Negative FHB
 - ⇒ Immunological test for pregnancy becomes negative.
 - ⇒ U/S show empty sac, no fetal cardiac motion & fetal movements

Management

- Nearly in 80% of cases spontaneous expulsion during expectant management can occur in three weeks' time.
- If immediate evacuation is decided, one has to use protocol for induced abortion

Septic abortion

- Abortion with clinical evidences of infection of the uterus and its contents.
 - ⇒ Fever at least 100.4°F (38°C) for 24 hours or more
 - ⇒ Offensive or purulent vaginal discharge
 - ⇒ Lower abdominal pain and tenderness.
- About 10% of abortions requiring admission to hospital are septic.
- Majority of are associated with incomplete abortion.

Cause of septic abortion

- Infection is usually due to
 - ⇒ Staphylococcus aureus
 - ⇒ Gram negative bacilli
 - ⇒ some Gram positive cocci.
 - ⇒ Mixed infections
 - ⇒ Anaerobic organisms
 - ⇒ Fungi
- The infection may spread, leading to salpingitis, generalized peritonitis, and septicemia.

Septic abortion...

Symptom	Signs
• Pyrexia • Chills, fever, sweating • Tachycardia • General malaise • Lower abdominal pain • Pelvic tenderness • Prolonged bleeding • Foul smelling vaginal discharge	• Distended abdomen • Tenderness • Tachycardia • Hypotension

Septic abortion...

Clinical grading	
Grade I	• The infection is localized in the uterus. • The commonest • Is usually associated with spontaneous abortion
Grade II	• The infection spreads beyond the uterus to the parametrium, tubes and ovaries or pelvic peritoneum.
Grade III	• Generalized peritonitis and/or endotoxic shock or jaundice or acute renal failure. • Almost always associated with illegal induced abortion.

Management

- Secure IV line and put on intravenous fluids
- Put on broad spectrum antibiotic
 - ⇒ Clindamycin 600 mg IV QID and gentamicin 5 mg/kg IV daily.
 - ⇒ If clindamycin is not available, give ampicillin 2 g IV every six hours and gentamicin 5 mg/kg body weight IV every 24 hours.
- A dilatation and curettage (D&C) or misoprostol
- AntiD if mother is Rh negative blood
- Treat any complication

Recurrent abortion

- A sequence of three or more consecutive spontaneous abortion before 20 weeks.
- It may be primary or secondary (having previous viable birth).
- A woman procuring three consecutive induced abortions is not a habitual aborter.

Etiology

Genetic factors

Endocrine and metabolic

Infection

Anatomic abnormalities

Chronic maternal illness

Unexplained.

Diagnostic

1. History and physical examination
2. Cytogenetic analysis of both partners
3. Ultrasound, HSG, hysteroscopy or laparoscopy
4. Luteal phase endometrial biopsy, mid luteal phase progesterone
5. TSH level, serum prolaction level
6. Lupus anticoagulant activity, Anticardiolipin IgG/IgM, IgM
7. Anti - Nuclear Antibodies
8. CBC, FBS, VDRL
9. Infection screening

Management

- Treatment specifically directed for identified cause
 - ⇒ Use of progesterone
 - ⇒ Anti thrombotic medications
 - ⇒ Repair of anatomic anomalies
 - ⇒ Correction of endocrine abnormalities
 - ⇒ Treatment of infections
 - ⇒ Cervical cerclage
 - ⇒ Counseling & psychological support

Differential diagnosis of abortion

- Ectopic pregnancy in intact tube
- Cervical polyp
- Vaginitis
- Cervical carcinoma
- Gestational trophoblastic disease
- Trauma
- Foreign bod

Induced abortion

- Is surgical or medical termination of a live fetus that has not reached viability.

Indications

- Maternal health problems like:
 - ⇒ Persistent cardiac decompensation
 - ⇒ Pulmonary arterial hypertension
 - ⇒ Advanced hypertensive vascular disease
 - ⇒ Diabetes with end-stage organ failure
 - ⇒ Malignancies
 - ⇒ Maternal mental illness
 - ⇒ Rape or incest
- Non compatible fetal with life
- Age of woman less than 18 years

Techniques for first-trimester

Surgical	Medical
• Dilatation and curettage • Vacuum aspiration • Menstrual aspiration	• Prostaglandins E2, F2α, E1, and analogues • Antiprogesterones—RU-486 (mifepristone). • Methotrexate—intramuscular and oral. • Various combinations of the above.

Complication of abortion

- Hemorrhage, shock, anemia
- Infection
- Intrauterine synechiae (Asherman's syndrome)
- Perforation of the uterus
- Injury to bowel and bladder
- Fistula formation
- Tetanus
- Ectopic pregnancy in later life
- Death

Post abortion care (PAC)

- Emergency treatment of incomplete abortion and potentially life threatening complications
- Post-abortion family planning counseling and services
- Links between post-abortion emergency services and the reproductive health care system.
- Community service provider partnership
- Counseling

Unit-8-Common gynecologist problem

Amenorrhea

Dysmenorrhea

Infertility

Menopause

Amenorrhea

Amenorrhea	• Is the absence of a menstrual period in a woman of reproductive age.
Primary Amenorrhea	• No period by age 13 in the absence of secondary sexual characteristics. • No period by age 15 plus presence of secondary sexual characteristics. • Approximately 2.5% of the population.
Secondary Amenorrhea	• No period 3 of the previous cycle intervals in oligomenorrheic women • No period 6 months of regular cycles • From 3% in the general population to 100% under conditions of extreme physical or emotional stress.

Causes of primary amenorrhea

Compartment 1- disorders of outflow tract	Congenital	Acquired(unresponsive endometrium)
	• Imperforate hymen • Transverse vaginal septum • Vaginal agenesis • Absent uterus • Absent vagina • Androgen insensitivity syndrome	• Asherman syndrome • Dilatation and curettage • Infection (tuberculosis) • Cervical stenosis
Compartment 2(problems in the ovary)	• Gonadal agenesis/ dysgenesis • Premature ovarian failure • Polycystic ovarian syndrome • Resistant ovarian syndrome • Some ovarian tumors	
Compartment 3(abnormality of pituitary gland)	• Pituitary tumors • Sheehan's syndrome(ischemic necrosis of pituitary gland 2ry to PPH)	
Compartment 4(abnormalities in hypothalamus)	• Stress, exercise • CNS infections-Tb • Sarcoidosis • Tumors • Kallmann's syndrome(congenital absence of GNRH) • Anorexia nervosa	

Causes secondary of amenorrhea

Common cause	⇒ Premature ovarian failure ⇒ Hyperprolactinemia ⇒ Pregnancy ⇒ Hypothalamic amenorrhea ⇒ Galactorrhea -amenorrhea syndrome ⇒ Androgen disorders • Polycystic ovarian syndrome • Congenital adrenal hypreplasia
Less common	⇒ Premature ovarian failure ⇒ Asherman's syndrome ⇒ Sheehan's syndrome
Rare	⇒ Diabetes ⇒ Hyperthyroidism or hypothyroidism ⇒ Cushing's syndrome or Addison's disease ⇒ Cirrhosis ⇒ Tuberculosis ⇒ Malnutrition ⇒ Irradiation or chemotherapy ⇒ Surgery

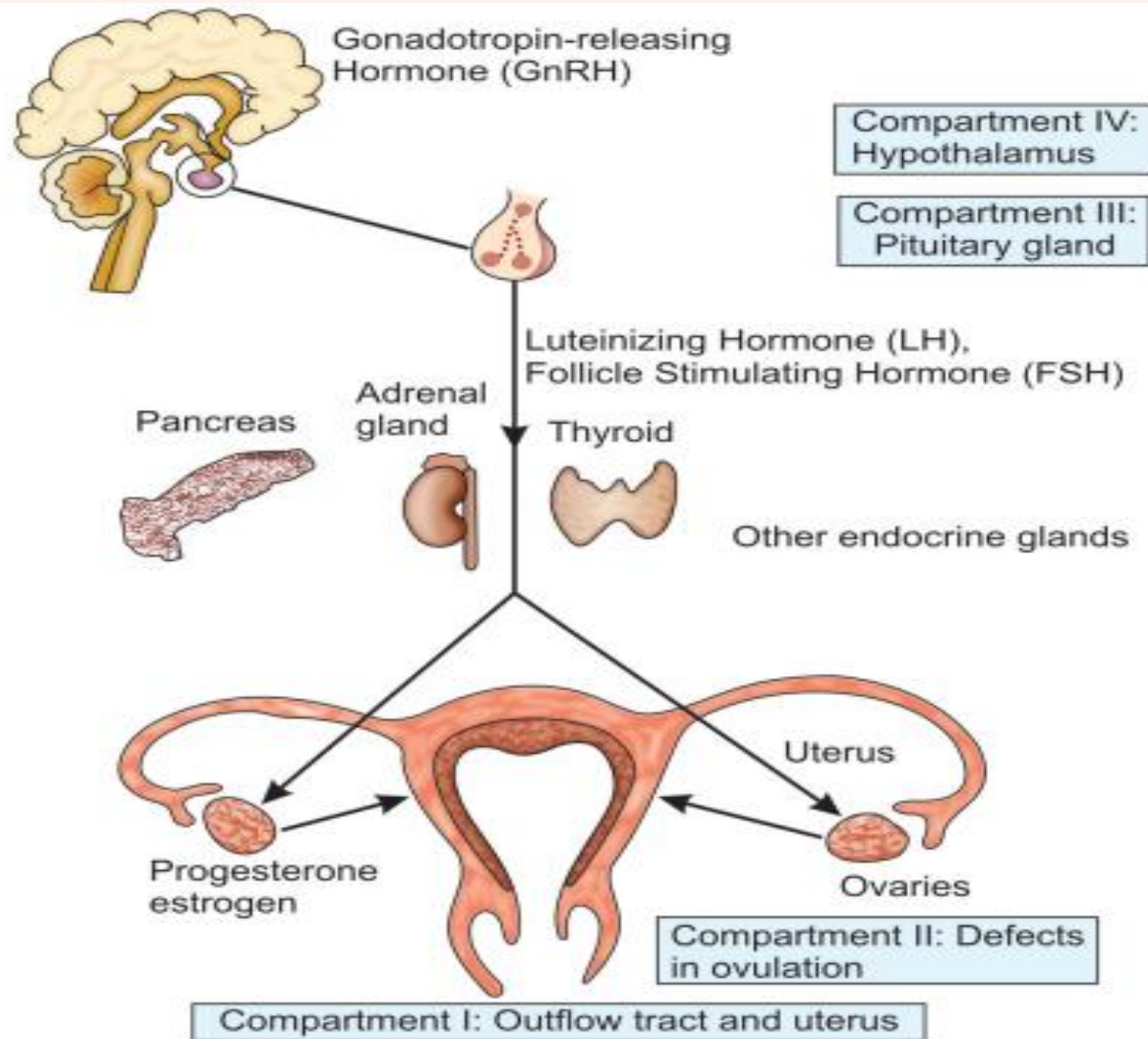


Fig. 27.1: Causes of primary amenorrhea related to defects in either of four compartments

Approach to amenorrhea patient

History and physical examination	Investigation
⇒ Age ⇒ Parity ⇒ Previous menstrual history ⇒ Mode of onset-Sudden, Gradual ⇒ Family history ⇒ Past medical history or recent illnesses ⇒ Weight fluctuation ⇒ History of any stressful events ⇒ History of drug intake ⇒ Radiation exposure ⇒ History of uterine curettage or uterine surgeries ⇒ History of PPH or shock or infection	• Human chorionic gonadotropin (hCG) • FSH and LH • Thyroid-stimulating hormone (TSH), • Prolactin (PRL) measured • Karyotype • Ultrasound

Treatments

- Treatments vary based on the underlying condition.
 - ⇒ Surgical correction if appropriate
 - ⇒ Estrogen therapy if oestrogen levels are low.
 - ⇒ Psychological treatment

Dysmenorrhea

- A crampy cyclic pain associated with menstruation.
- Affecting **>50-60%** of menstruating women
- The pain is colicky in nature.
- Classified as
 1. Primary
 2. Secondary on the basis of associated anatomic pathology.

Primary dysmenorrhea

- Recurrent, crampy lower abdominal pain that occurs during menstruation in the absence of pelvic pathology.
- It is the most common gynecologic complaint among adolescent females.
- Affects 60 to 93% of adolescent females
- Usually appears within 1 to 2 years of menarche, when ovulatory cycles are established.

Pathophysiology of primary dysmenorrhea

- Excessive or imbalanced amount of **prostaglandins** secreted from the endometrium during menstruation.
 - ⇒ **Increased uterine contractions with a dysrhythmic pattern**
 - ⇒ **Increased basal tone and increased active pressure.**
- Uterine hypercontractility, decreased uterine blood flow, and increased peripheral nerve hypersensitivity contribute to pain.

Symptom of primary dysmenorrhea

History	• Age at menarche • Duration of menstrual cycles • Interval between menstrual periods • Date of last two menstrual periods • Onset and duration of cramps • Nausea, vomiting back pain, dizziness, or headache • Medication use • Ability to engage in all daily activities • Sexual history
Signs	• The vital signs are normal • Suprapubic region may be tender to palpation. • Bowel sounds are normal • No upper abdominal tenderness • No abdominal rebound tenderness.

Treatment of primary dysmenorrhea

- For severity of menstrual pain and limitation of daily activities
- Patient reassurance and education.
- Hormonal contraceptive agents(COC, POP)
- NSAIDs
 - ⇒ Ibuprofen, 400 mg, P.O., TID OR
 - ⇒ Mefenamic acid, 500mg, P.O, TID.
 - ⇒ Prior to the onset of menses or at the onset of pain every 6 to 8 hours for the first few days of menses.

Secondary dysmenorrhea

- Cyclic menstrual pain that occurs with underlying pelvic pathology.
- Usually develops years after menarche
- Can occur with anovulatory cycles.
- Often begins 1 to 2 weeks before menstrual flow and persists until a few days after the cessation of bleeding.

Cause of secondary dysmenorrhea

- The most common cause is
 1. Endometriosis-most common
 2. Adenomyosis
 3. IUCD- nonhormonal
 4. Infection
 5. Ovarian cysts
 6. Congenital pelvic malformations
 7. Cervical stenosis.

Treatment

- The specific pathologic cause
- The treatment of the underlying disorder
 - ⇒ Endometriosis
 - ⇒ Adenomyosis
 - ⇒ PID

Infertility

- Is inability to conceive after 1 year of unprotected intercourse of reasonable frequency.
- Affecting 10 to 15 % of reproductive-aged couples.
- Subfertile, as they will ultimately conceive if given enough time.
- Sterility: incapable of becoming pregnant or of inducing pregnancy.

Infertility....

- Fecundability is the ability to conceive in a single cycle
 - ⇒ The monthly probability of conceiving is 20 to 25 %.
 - ⇒ Approximately 50 % of women will be pregnant at 3 months
 - ⇒ 75 % will be pregnant at 6 months
 - ⇒ ≥ 85 % will be pregnant by 1 year.

Etiology of infertility

- Successful pregnancy requires
 - ⇒ Patent out flow tract
 - ⇒ Ovulation
 - ⇒ Sperm of adequate number & quality
 - ⇒ Ovum pick-up by a fallopian tube
 - ⇒ Fertilization
 - ⇒ Transport of a fertilized ovum into the uterus
 - ⇒ Receptive endometrial cavity.
- Infertility can be seen as male & female infertility

Etiology of infertility

- The major causes of infertility in female could be
 - ⇒ Fallopian tube occlusion or anovulation.
- In males, it could be pre-testicular, testicular or post-testicular affecting the semen quality in terms of volume, sperm count, morphology, concentration, and motility

Etiology of infertility

Cause	Description
Infection:	• STI (chlamydial, gonorrhea, syphilis etc.) • Infectious and parasitic diseases: mumps, TB, schistosomiasis • Post-partum/abortal
Exposure to toxic substances	• Arsenic, aflatoxins, pesticides, caffeine, tobacco, alcohol, heat
Anovulation	• The main cause being polycystic ovarian syndrome (PCOS) in 75%of cases
Surgery, trauma	

Etiology of male infertility

Etiology of infertility	Site of dysfunction	Prevalence %
Pre-testicular	Hypothalamic / pituitary	1 -2
Testicular	Testis	30 - 40
Post-testicular defects	Ducts of sperm transport	10 - 20
Idiopathic	-----	40 -50

Clinical features

- History of STD, features of endocrinopathies,
- History of contraception usage, surgical history (ovarian cyst, pelvic abscess, orchidopexy, repair of inguinal hernia)
- Medicine intake, diet (smoking, alcohol and caffeine)
- Galactorrhea
- Menstrual pattern, dysmenorrhea
- Current illness
- Occupation (male)

Investigations

- CBC
- FBS
- Vaginal smear
- Hormone profile:
 - ⇒ Serum FSH and LH
 - ⇒ Serum progesterone
 - ⇒ TSH, T3, T4, Testosterone, DHEA, DHEAS, prolactin
- Hysterosalpingography (HSG)
- Ultrasound: ovarian volume, counting ovarian follicles
- Laparoscopy: to look for endometriosis and tubal patency
- For male: Sperm analysis, vasography, trans-rectal ultrasound, hormone assay (FSH, LH, prolactin, thyroid function tests, testosterone)

Diagnostic work up of male infertility

Ser No	Normal semen analysis findings	
1	Volume	≥ 1.5 mL
2	Count	≥ 20 million/mL
3	Motility	$\geq 50\%$
4	Morphology	$\geq 30\%$
5	WBCs	≤ 1 million/mL
6	PH	7.2-7.8

Terminologies in seminal fluid analysis

- **Aspermia**: failure of formation or emission of semen
- Oligospermia: < 20 million/ml
- Polyzoospermia: > 350 million/ml
- **Azoospermia**: no spermatozoa
- Asthenozoospermia: **abnormal** motility
- Teratozoospermia: **abnormal** morphology
- Necrozoospermia: dead or motionless

Evaluation of infertility

- Objectives:

- ⇒ Discover etiologic factor

- ⇒ Rectify the abnormality to improve fertility

- ⇒ Reassure the couples

- When to investigate

- ⇒ After one year

- ⇒ After 6 months in women older than 35 years & after 40 years in men

Evaluation of infertility

History	Examination	Investigation
• Childhood illnesses • Exposure to toxins, heat, radiation, metals • Infection, trauma, torsion, DES exposure • Precocious or delayed puberty • Galactorrhea • Change in the pattern of hair distribution • Drug exposure • Medical illnesses • Surgical illnesses & their treatment	• Features of hypogonadism secondary sexual characteristics eunuchoidal skeletal proportions, hair distribution. • Testes evaluation: size, weight, volume • Epididymis: irregularity • Visual field loss • Spermatic cord :vas deferens, varicose • Prostate: size, tenderness • Penile abnormalities • Breast, liver, thyroid, neurological evaluation	• Menstrual pattern • Endometrial biopsy • Serum progesterone • Sonography • Laparoscopic visualization of recent corpus luteum • Detection of ovum from aspirated peritoneal fluid

Treatment of the Infertile couple

Lifestyle Therapies	• Weight optimization • Stop smoking • Exercise • Stress management
Correction of an Identified Cause	• Ovulation Induction for ovulatory dysfunction • Surgical correction of tubal ,peritoneal & uterine defects • Surgical and medical treatment of male infertility
Assisted Reproductive Technologies	• In vitro fertilization (IVF) • Intracytoplasmic sperm injection (ICSI) • Egg donation • Gestational carrier surrogacy • Gamete intrafallopian transfer (GIFT) • Zygote intrafallopian transfer (ZIFT)

Menopause

Term	Definition
Menopause	• A 12 months of amenorrhea in women over age 45 years in the absence of biological or physiological causes
Pre-menopause	• the period before menopause during which menses is irregular and climacteric symptoms are experienced
Post-menopause	• Period after menopause
Peri menopause	• Pre and post menopause (40-55 years)
Premature ovarian failure	• Menopause prior to age 40 years
Late menopause:	• If menopause occurs after age 55 years

Age at menopause

- Average age at menopause is ~ 51yrs
- Late menopause: accounts for 5% of women, after age 55yrs
- Early menopause accounts 5%, b/n ages 40 - 45yrs.

Factors that affect age at menopause

Genetics	• Genetically programmed loss of follicles • Genes for apoptosis • Genetic variations in estrogen receptors • Family history of age menopause
Menses	• Age of menarche – no correlation • Shorter cycle length during adolescence – earlier menopause • Irregularity in early 40s – earlier menopause
living in high altitude	• * Early menopause
Smoking	• ** Earlier menopause
Growth restriction	• Earlier menopause
Alcohol	• Later menopause
Race and Parity	• No influence

Pathophysiology

- Ovarian follicles become depleted and resistant to gonadotropin hormones.
- Estradiol production diminishes which result in high level of FSH and LH .
- when follicles are absent to be stimulated estrogen level falls causing hypoestrogenic state that follows menopause leading to medical conditions associated with climacteric.

Changes in menopause

- Hormonal changes
 - ⇒ Increased FHS and LH levels ($>40\text{iu/lit}$) for 2-3 years then remain stable or slightly decrease. $\text{FHS} > \text{LH}$
 - ⇒ Estradiol levels decrease ($<20\text{pg/liter}$)
 - ⇒ Progesterone levels are low
 - ⇒ Levels of androgens fall
- Menopause/post menopause $\Rightarrow \text{FSH}(20\times) / \text{LH}(3\times)$

Organ changes

- Uterus
- Ovary: shrinks & white
- Vagina & vulva
- Breasts
- Bone
- Bladder & urethra

Reproductive organs

- Vaginal atrophy (epithelial thinning and flattening of rugae)
- Cervical atrophy with shallow fornices, decreased cervical mucus
- Decreased uterine size ,endometrial atrophy
- Decreased impalpable ovaries
- Supporting muscles lose their tone
- The labia lose fat and flatten

Other organs

- Bladder and urethral epithelium atrophy
- Decreased size of breasts
- Thining and loss of elasticity of skin
- Bone loss

Clinical manifestations

- Bleeding patterns: oligomenorrhea $\Rightarrow$ amenorrhea
- Vasomotor symptoms: hot flush, episodic vasomotor disturbance with sudden flushing followed by sweating affects 70% women & only 20% need help.
- Hot flushes (burning in face, neck and chest with sweating)
- It is self limited and usually resolves 1-5 years after menopause.
- Sleep disturbance: primary sleep disorder or arousal problem.

Clinical manifestations

- Genitourinary:
 1. Vaginal dryness: atrophic vaginitis
 2. Sexual dysfunction:
 3. Vaginal dryness & dyspareunia
 4. Other urinary symptoms: include atrophic urethritis, diminished urethral mucosa, incontinence, & recurrent UTI
- Depression: association between menopausal transition & depression

Clinical manifestations

- Skin changes: collagen content of the skin & bone decreases
- Long term issues:
 - ⇒ Osteoporosis
 - Treated with estrogen replacement.
 - Prevented by estrogen, exercise and calcium intake
 - ⇒ Atherosclerotic disease of the heart
 - ⇒ Dementia

Investigations

- Hormone tests if available (serum LH, FSH, estradiol).
- Routine investigations e.g. CBC, blood glucose, lipid profile.
- X-ray to evaluate bone density

Management

- Non pharmacologic
 - ⇒ Counseling and reassurance.
 - ⇒ Encourage active lifestyles, exercise and regular physical checkups for common medical problems.
 - ⇒ Avoid hot weather conditions
 - ⇒ Light clothing, cold shower
 - ⇒ Balanced die

Management

- Hormonal replacement therapy with indication
 - ⇒ Estrogen deficiency symptoms(vaginal dryness, dyspareunia, vasomotor symptoms)
 - ⇒ Osteoporosis
 - ⇒ Premature ovarian failure
 - ⇒ Surgical menopause

Unit-9:Malformations of the female genital tract

Uterine anomalies

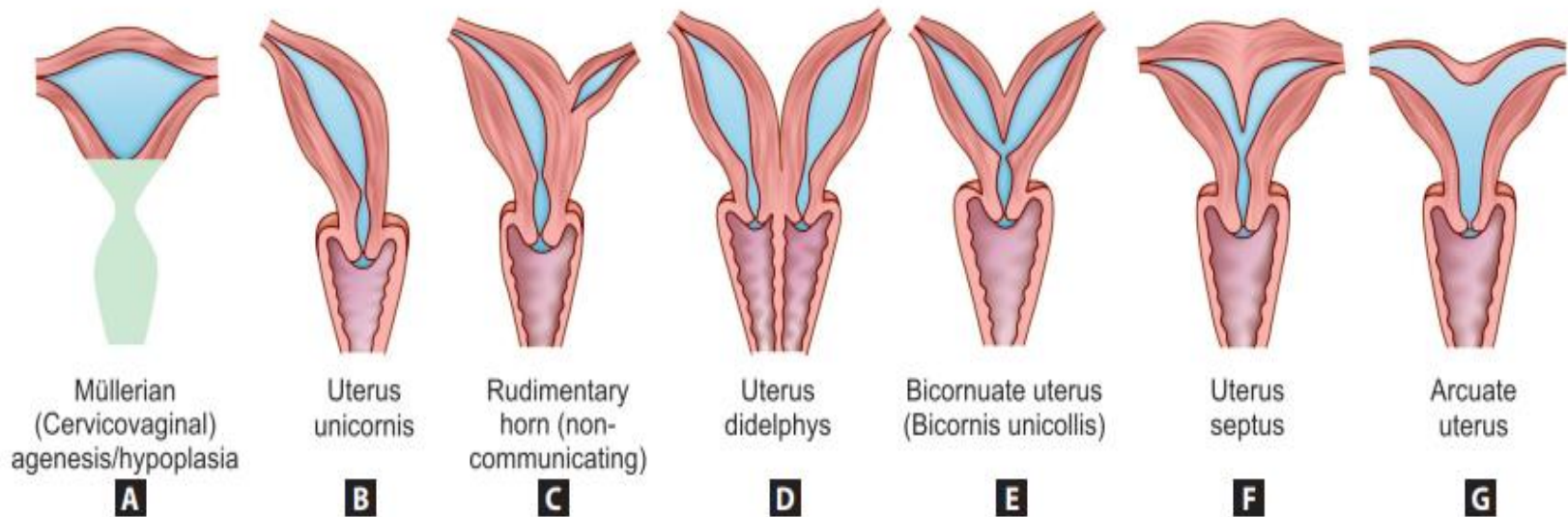
Fallopian anomalies

Vaginal anomalies

Uterine anomalies

Class	Uterine anomalies	Description
I	Müllerian agenesis	Absence of vaginal, cervical, uterine, tubal
II	Unicornuate uterus	
III	Uterus didelphys	Double uterus, double cervix and a double vagina
IV	Bicornuate uterus	
V	Septate Uterus	
VI	Arcuate uterus	
VII	Diethylstilbesterol (DES)-related anomalies	Varieties of malformations (vagina,uterus, fallopian tube)

Uterine anomalies...



Figs 4.5A to G: Classification of Müllerian anomalies

Clinical features

- Infertility and dyspareunia
- Dysmenorrhea
- Menstrual disorders
- Midtrimester miscarriage
- Rudimentary horn pregnancy
- Cervical incompetence.
- Increased incidence of malpresentation
- Preterm labor, IUGR, IUD.
- Prolonged labor
- Obstructed labor
- Retained placenta and postpartum hemorrhage

Diagnosis

- History
- Physical examination
- Imaging
 - ⇒ Hysteroigraphy
 - ⇒ Hysteroscopy
 - ⇒ Laparoscopy
 - ⇒ Ultrasonography (vaginal probe)
 - ⇒ MRI

Reproductive outcome

- Better obstetric outcome in septate uterus (86%), bicornuate uterus (50%) has been mentioned.
- Unicornuate uterus has very poor (40%) pregnancy outcome.
- No treatment is generally effective.
- Uterus didelphys has best possibility of successful pregnancy (64%).

Abnormalities of the fallopian tubes

- The tubes may be unduly elongated
- The tube may have accessory ostia or diverticula.
- Rarely, the tube may be absent on one side.
- These conditions may lower the fertility or favor ectopic pregnancy.

Vaginal anomalies

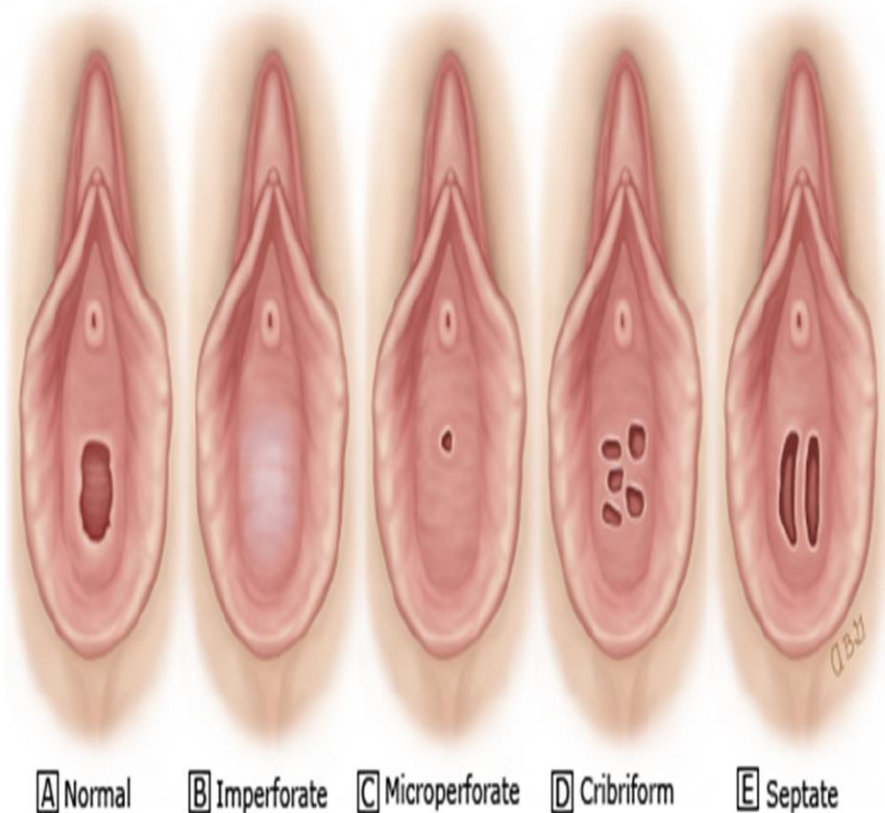
Imperforate hymen

Transverse or longitudinal vaginal septum

Vaginal atresia (agenesis of the lower vagina)

Imperforate hymen

- Is formed during fetal development when the sinovaginal bulbs fail to canalize with the rest of the vagina.
- Patients has
 - ⇒ Hydro/ hemato/ mucocolpos
 - ⇒ Primary amenorrhea
 - ⇒ Cyclic pelvic pain.
 - ⇒ Marked distension of the vagina



Imperforate hymen...

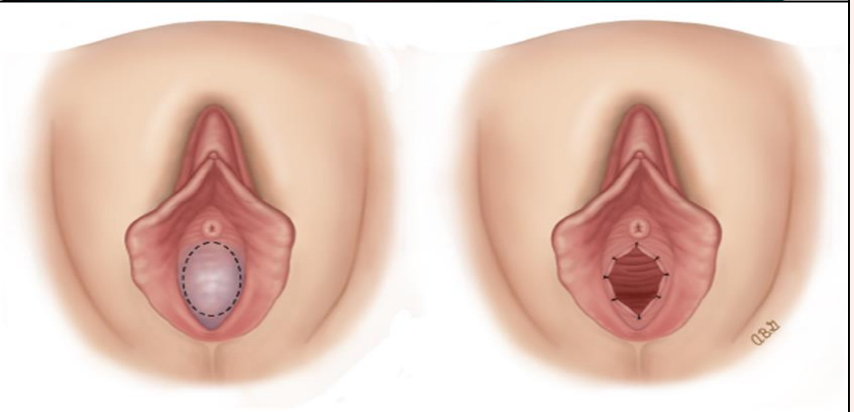
- Physical examination
 1. Absence of vaginal lumen
 2. Tense bulging hymen a bluish discoloration
 3. Possibly increasing lower abdominal girth(hematometra).



Fig. 4.3: Tense bulging of the hymen in hematocolpos

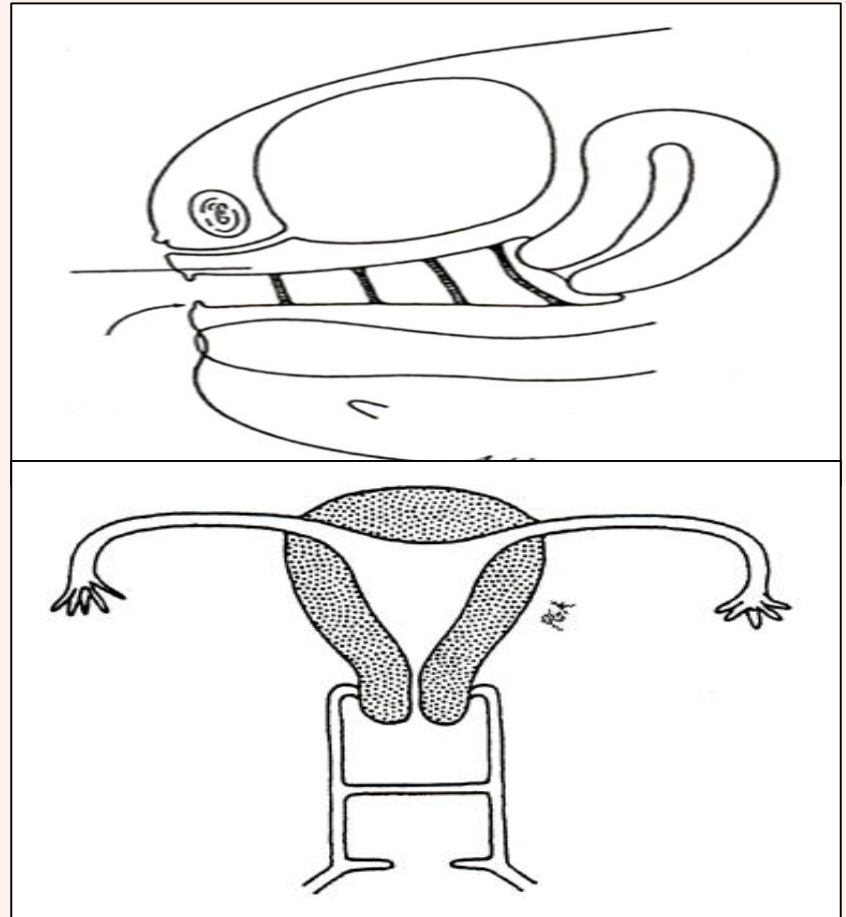
Management

- OCP taken continuously to suppress the menstrual cycle
- NSAIDs to relieve pain.
- Surgical intervention in adolescents



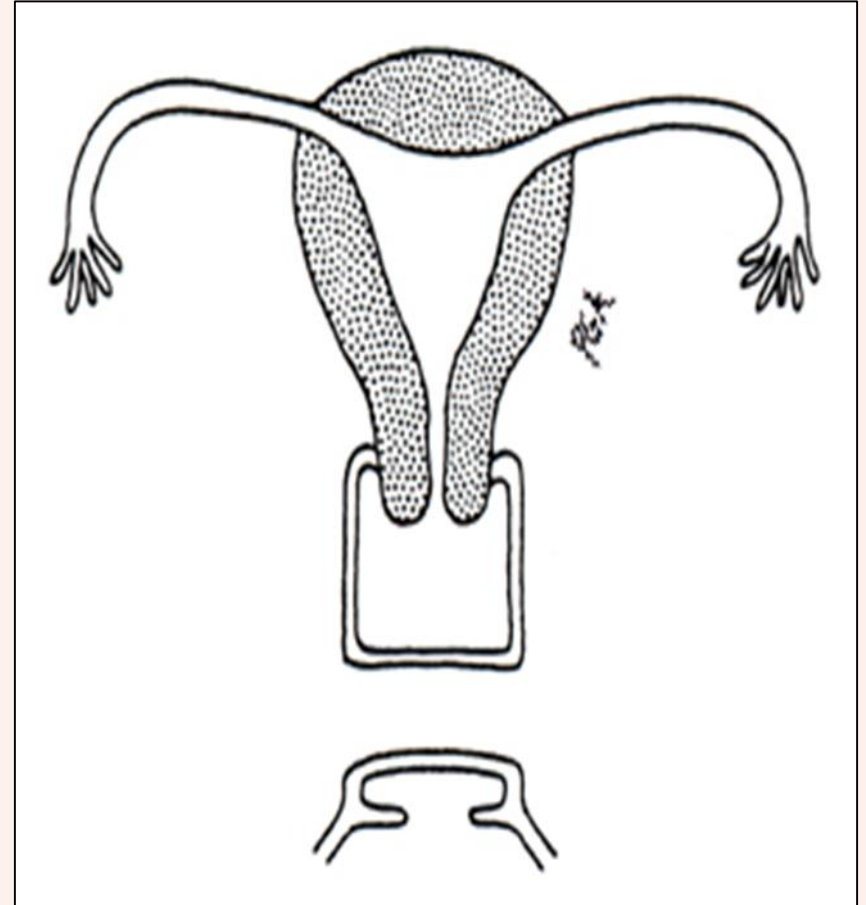
Transverse or longitudinal vaginal septum

- Is failure off fusion and/or canalization of the urogenital sinus and müllerian ducts.
- Treatment a small septum can be resected, followed by an end-to-end anastomosis of the upper and lower vaginal mucosa.



Vaginal atresia (agenesis of the lower vagina)

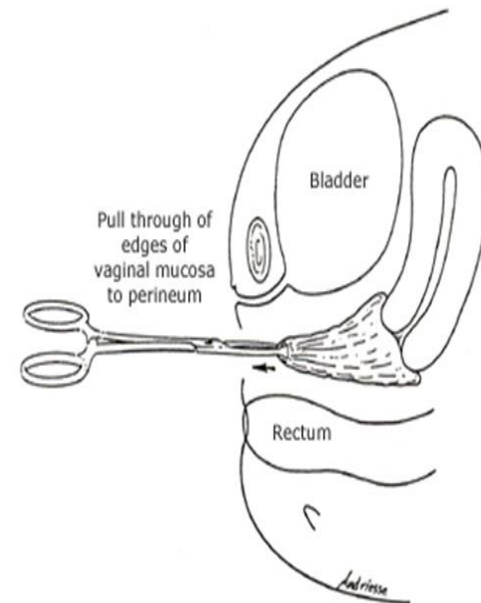
- When the lower vagina fails to develop and is replaced by fibrous tissue
- Results when urogenital sinus fails to contribute the lower portion of vagina.
- It presents during adolescence with
 1. Primary amenorrhea
 2. Cyclic pelvic pain



Treatment

- Treatment involves a combination of
 - ⇒ Psychosocial support
 - ⇒ Counseling
 - ⇒ Nonsurgical
 - ⇒ Surgical correction (vaginal pull-through procedure.)

Vaginal pull through procedure



After drainage of obstructed lower vagina, the normal upper vaginal tissue is pulled through to the perineum to create a normal vagina.

Unit -10: Benign and malignant tumors of genital tract

- Objective

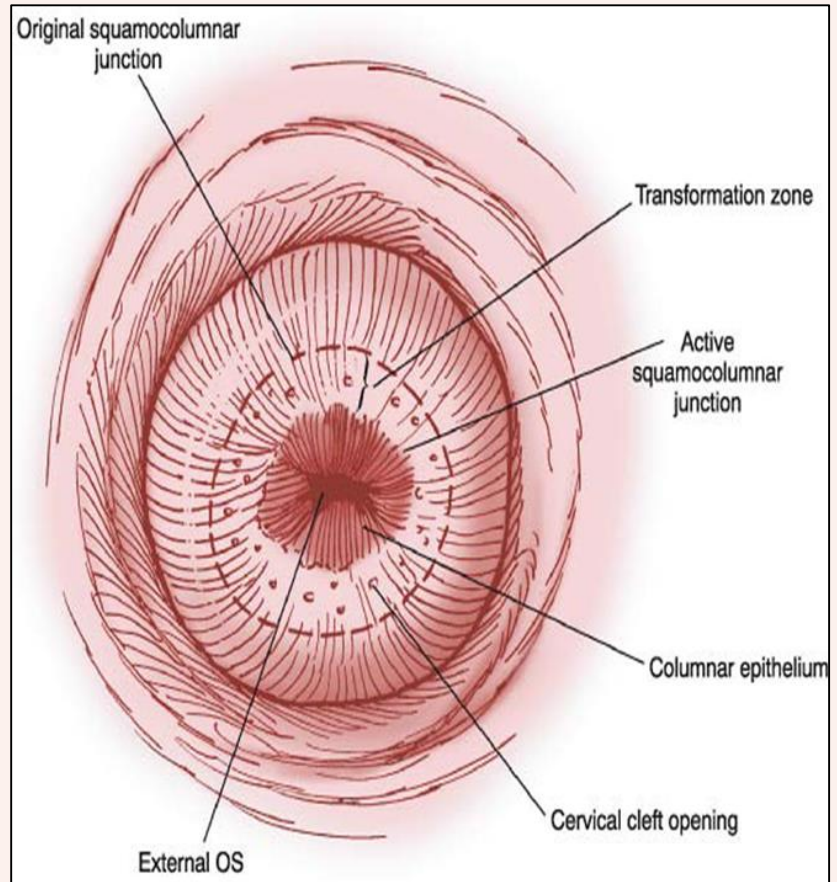
⇒ Explain risk, clinical presentation, diagnosis and treatment
benign and malignant tumors of the genital tract & breast

Cervical cancer

- Is the second most common cancer in the world
- The third common cause of deaths from cancer in women.
- Diagnosed case each year, an estimated 530,000
 - ❖ More than 270,000 women die from it.
 - ⇒ 85% occur in developing countries.

Origin of cancer

- Metaplasia advances from the
 - ⇒ Original SCJ inward
 - ⇒ Toward the external os
 - ⇒ Over the columnar villi establishes at transformation zone



Risk factors

Human papilloma virus 16/18

Early sexual activities (cohabitarche)

Multiple sexual partners

Partner with multiple sexual partner

Genital hygiene

Smoking

Infection

Immunosuppression (HIV/AIDS)

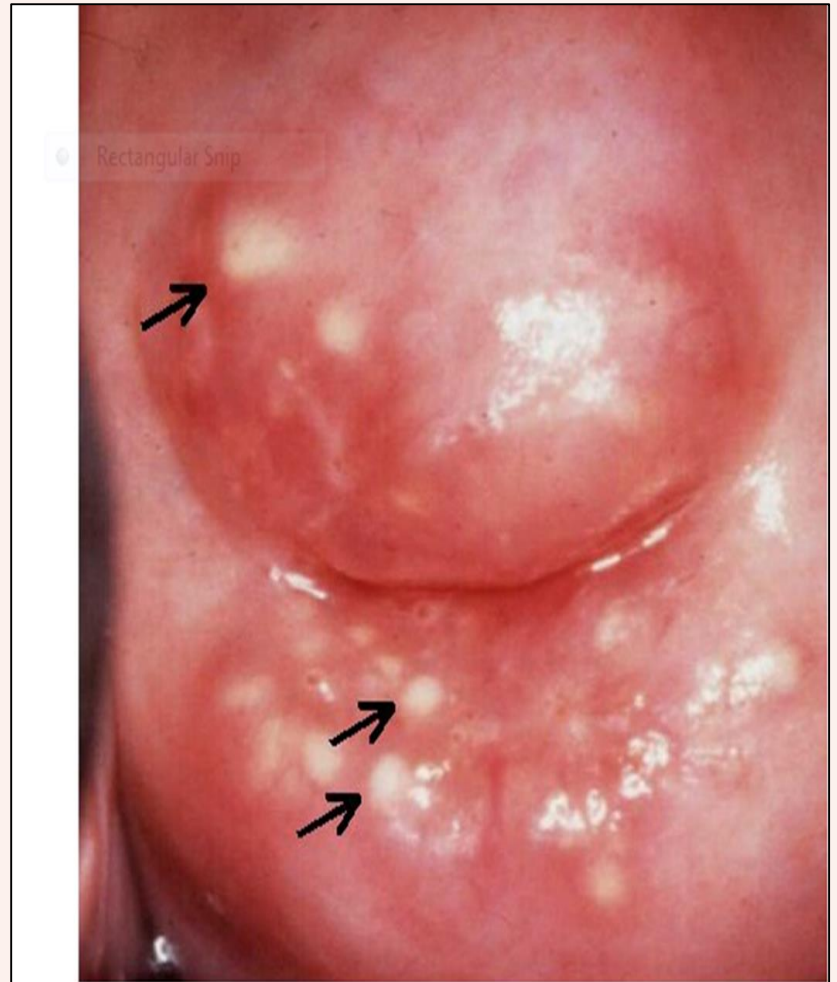
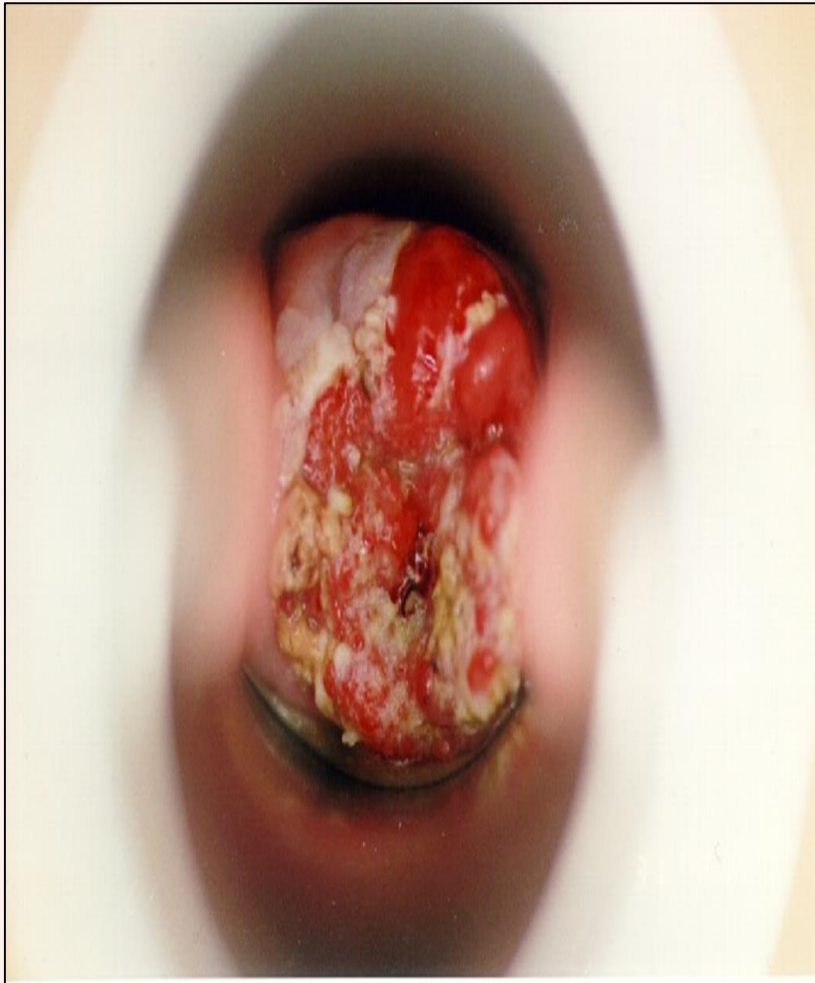
Cervical cancer staging

Stage	Characteristics
0	• Carcinoma in situ, cervical intraepithelial lesion (CIN)
I	• Carcinoma is strictly confined to cervix
II	• Carcinoma extends beyond cervix plus 2/3 upper vaginal • No pelvis side wall
III	• Carcinoma has extended to the pelvic wall • No cancer-free space b/n tumor and pelvic wall(PR) • Tumor involves lower third of vagina • Hydronephrosis or nonfunctioning kidney
IV	• Carcinoma has extended beyond true pelvis • Involved mucosa of bladder or rectum

Clinical presentation of invasive cervical cancer

Stage Ia:	Stage Ib, II	Stage III, IV
• None • Asymptomatic	• Bleeding abnormalities • Vaginal discharge	• Back pain • Lymph edema of the legs • Dysfunction of rectum (bloody stools, subileus) • Hematuria • Dysuria • vagina and vulva are often moist and bloody

Clinical presentation of invasive cervical cancer..



Diagnosis

- Blood (CBC ,ESR ,Serum electrolytes Coagulation status)
- OFT(BUN and Creatinine, SGOT, SGPT, alkaline phosphatase)
- Urinalisys
- Blood glucose level
- Tumor marker (Squamous cell cancer antigen and CA 12-5).
- Papanicolaou smear⇒Histologic evaluation of **cervical biopsy**
- Colposcopy and cervical biopsy

Management principle

- Supportive care
- Operation and medical management
 - ⇒ Operative treatment(Stage Ia,Ib, IIa and IIb)
 - ⇒ Radiotherapy of cervical cancer(FIGO III-IV)
 - ⇒ Chemotherapy of cervical cancer(FIGO III-IV)
- Palliative care

Complication

1. Hemorrhage
2. Infections
3. Peritonitis
4. Anemia
5. Uremia
6. Death

Prevention

Primary prevention	• Girls 9-14 years -HPV vaccination • Health information and warnings about tobacco use. • Sexuality education tailored to age & culture. • Condom promotion/provision for those engaged in sexual activity. • Male circumcision.
Secondary prevention	• Women > 30 years of age • Screening and treatment as needed • “Screen and treat” with low cost technology VIA followed by cryotherapy • HPV testing for high risk HPV types
Tertiary prevention	• Treatment of invasive cancer at any age ⇒ Ablative surgery ⇒ Radiotherapy ⇒ Chemotherapy

Uterine neoplasms

Benign	Malignant
• Polyp • Adenomyosis • Myoma	• Primary ⇒ Endometrial Cancer ⇒ Sarcoma • Secondary arise from ⇒ Ovaries ⇒ Breast ⇒ Bowel

Uterine leiomyomas

- Called myomas, fibroids, fibromyomas
- Are the most common pelvic benign tumor in women
- Are proliferative, well circumscribed, pseudocapsulated benign tumors composed of smooth muscle and connective tissue.
- They arise in reproductive-age women

Uterine leiomyomas

- Are the commonest benign tumor of the uterus
- Are more common in nulliparous or having one child
infertility
- Their size ranges from 1mm to 20cm or more
- May be single or multiple

Uterine liomyomas...

- Can occur out side the uterus

- ⇒ Fallopian tubes

- ⇒ Vagina

- ⇒ Round ligament

- ⇒ Uterosacral ligament

- ⇒ Vulva

- ⇒ Gastrointestine

Predisposing factors

- Risk factors are

- ⇒ Nulliparity

- ⇒ Obesity

- ⇒ Hyperoestrogenic state

- ⇒ Black women

- ⇒ Alcohol

- ⇒ Hypertension

- Protective factor

- ⇒ Multiparity

- ⇒ Dietary vitamin

- ⇒ Smoking reduce the risk.

- ⇒ Progestin-only contraceptives

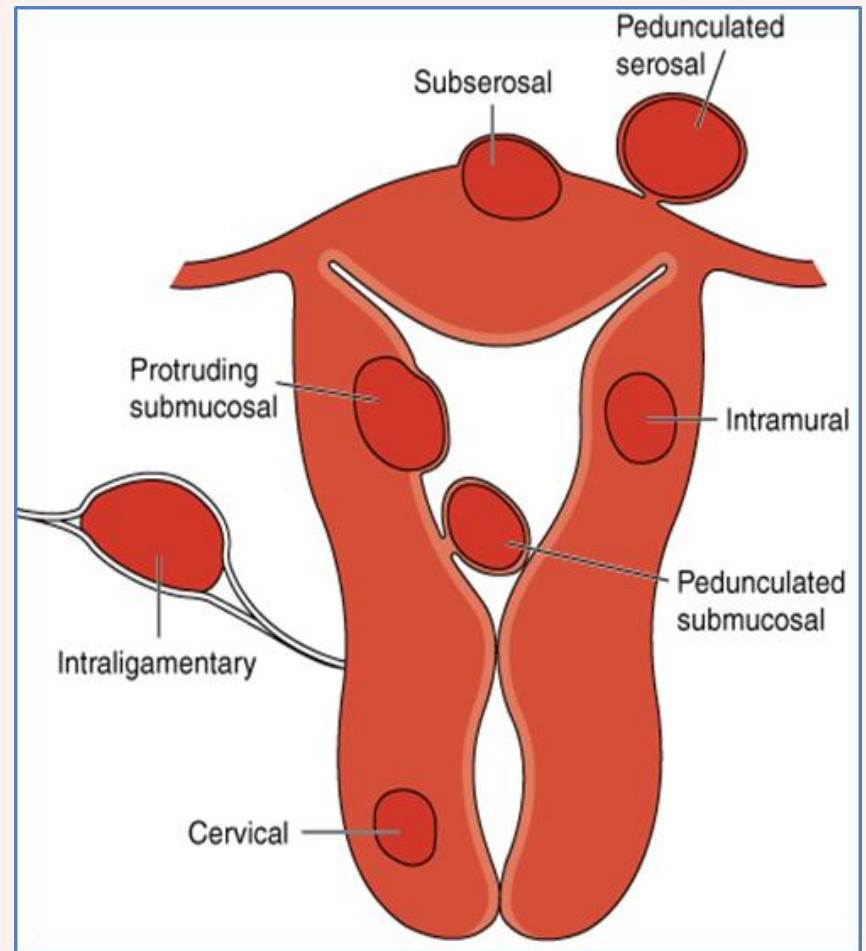
Types(according to location)

Intramural myomas

Subperitoneal or subserous

Submucous myomas

Cervical



Clinical features

Bleeding	Bulk-related symptoms	Reproductive dysfunction
• Heavy or prolonged menstrual bleeding. • 75% are asymptomatic • Most commonly menorrhagia (metrorrhagia)=heavy and prolonged • Menorrhagia-excessive bleeding during menses(more than80ml)	• Feeling of pelvic heaviness • Urinary frequency • Urinary retention • Ureteral obstruction • Constipation • Varicosity or edema • Pelvic pain: chronic, intermittent, dull pressure or pain.	• Ectopic pregnancy • Infertility due to • Distortion and elongation • Preventing rhythmic uterine contraction • Cornual block • Anovulation • Abortion • Preterm labor

Diagnostic evaluation

- History
- Physical examination
- Laboratory testing(Hgb,CBC,....)
- Pelvic U/S is first-line study used to evaluate for uterine fibroids.

Management

- Conservative for small, diagnosis is certain and no symptoms
 - ⇒ Patient may be reassured but kept under regular supervision
- Surgical
 - ⇒ Myomectomy with is need of further fertility
 - ⇒ Hysterectomy
- GNRH analogues
- Uterine artery embolization(UAE)
- Myolysis
- Immunotherapy

Endometrial cancer

- Most common genital tract malignancy in developed countries
- 50% of all gynecologic malignancies
- 4th most common malignancy in developed world next to breast,CRC,lunge
- The average age at diagnosis is the early 60s

Endometrial cancer risk factor

Factors Influencing Risk	Estimated Relative Risk ^a
Obesity	2–5
Polycystic ovarian syndrome	>5
Long-term use of high-dose menopausal estrogens	10–20
Early age of menarche	1.5–2
Late age of natural menopause	2–3
History of infertility	2–3
Nulliparity	3
Menstrual irregularities	1.5
Residency in North America or northern Europe	3–18
Higher level of education or income	1.5–2
White race	2
Older age	2–3
High cumulative doses of tamoxifen	3–7
History of diabetes, hypertension, or gallbladder disease	1.3–3
• Protective factors ⇒ Long term use of COC. ⇒ Cigarette smoking.	

Clinical features

- An average age of 60 year .
- Vaginal bleeding or discharge(90%)
- Pelvic pressure or discomfort(enlargement or extrauterine spread)
- Hematometra or pyometra(cervical stenosis)⇒purulent vaginal discharge
- <5% of women diagnosed are asymptomatic.
- Risk like obesity,hypertension

Clinical features...

- Peripheral lymph nodes and breasts should be assessed carefully
- Ascites or hepatic or omental metastases may be palpable.
- Speculum examination may confirm the presence of bleeding
- Atrophic vaginitis
- Enlarged, fixed and immobile uterus from parametrial extension.

Diagnosis

- CBC
- Endometrial biopsy is 90% to 98% accuracy
- Imaging
 - ⇒ Ultrasound
 - ⇒ Computer tomography
 - ⇒ Magnetic resonance imaging (MRI)

Histologic classification

Endometrioid Adenocarcinoma

Secretory carcinoma

Mucinous carcinoma

Papillary serous carcinoma

Clear cell carcinoma

Squama's cell carcinoma

Endometrial cancer clinical stage

Stage	Characteristic	Distribution
I	limited to endometrium	70
II	Extends to the glandular epithelium of the endocervix	13
III	Involves serosa and/or adnexa	14
IV	Invades bladder mucosa and/or bowel mucosa	3

Pretreatment evaluation

- A complete history
- Physical examination
- Chest radiography should be performed to exclude pulmonary metastasis
- CBC, BG and Rh, OFT,urinalysis,CA125
- Ultrasonography and MRI to assess myometrial invasion

Endometrial ca treatment

Surgery	• Abdominal Hysterectomy & BSO with cytology(>IB) • Primary operative procedures for carcinoma of endometrium. • Remove adnexa-the site of microscopic metastasis.
Radiation therapy	• Progestin (because most endometrial cancers have both E and P receptors)
Chemotherapy	• Adjuvant cytotoxic chemotherapy has been studied in a few trials • No benefit to adjuvant chemotherapy could be found

Reference

